Davide Curto

Portfolio

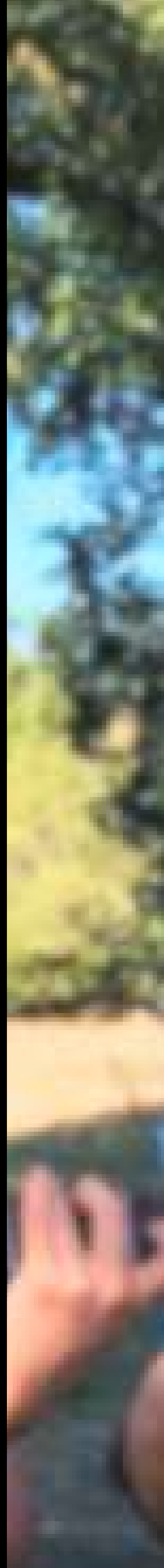
TRIVIO
COSTIERA AMALFITANA
Salerno, Italia, sud Europa
Scegli questa meta
Costo della vita
Clima
Target di età
Attività
COFFEE

Content

Work in progress

ABOUT ME

Hi, I'm Davide Curto, a designer graduated at the school of architecture at "Sapienza university of Rome".
I see design as a systemic language capable of integrating services, products and branding, into a single and cohesive experince. Guided by my value as a scout leader, I believe in the educational and well-being power of design.
I'm also a big fan of music, playing piano and guitar, just as a hobby.

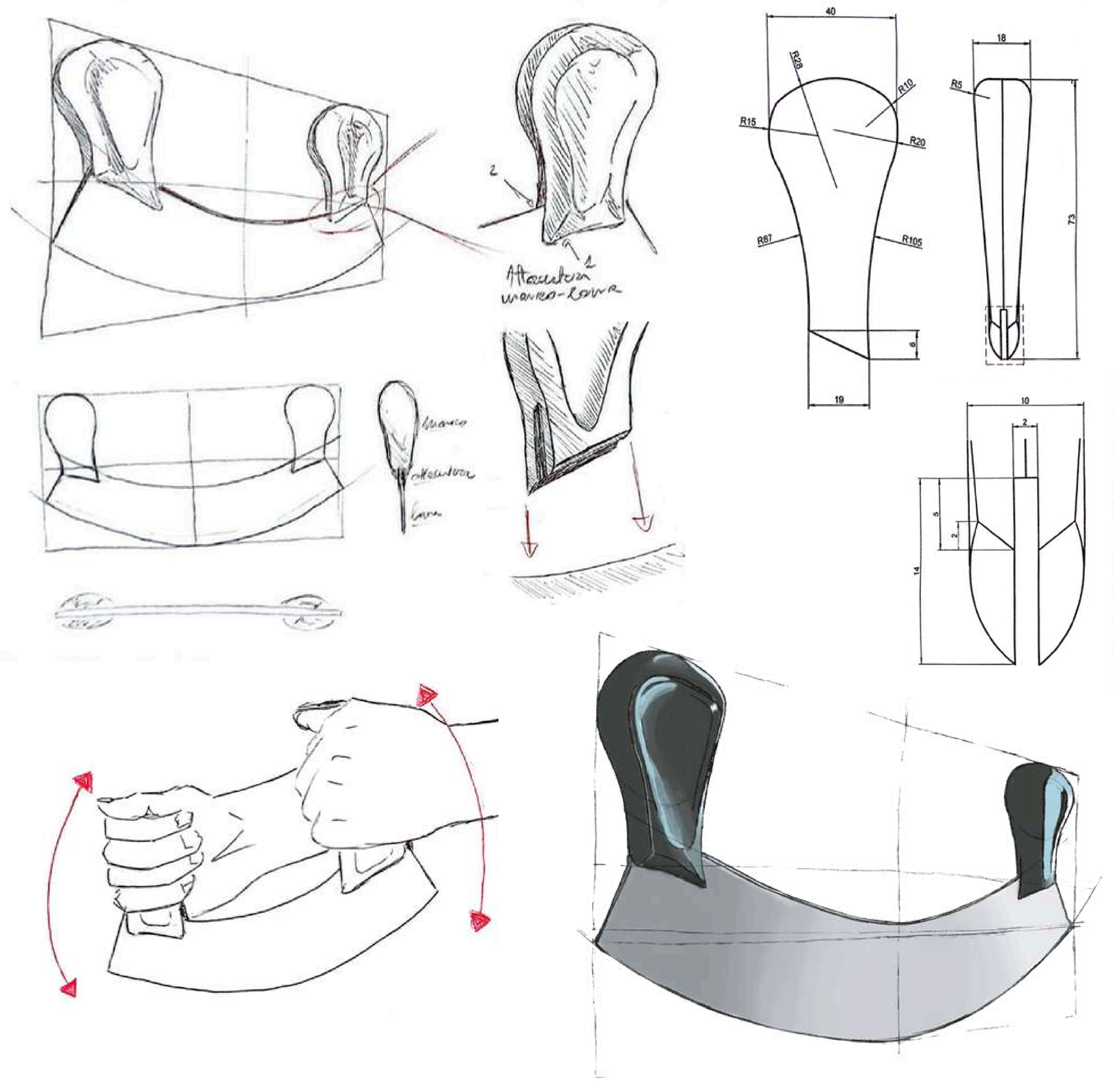


MESALUNA

Mezzaluna knife

My first university project. The brief involved the re-design of a kitchen utensil, for a famous Italian brand and realize the prototype and the design the exhibition layout in the 35x25x35 Ikea Eket cube. I selected the mezzaluna knife as the object and Ichnusa as the brand.

Product analysis



BRAND ANALYSIS

”

Ichnusa is a bottom-fermented beer with a clear golden color and a persistent, fine-grained, compact foam. Its aroma has a light hint of hops that gives it a pleasantly bitter taste.

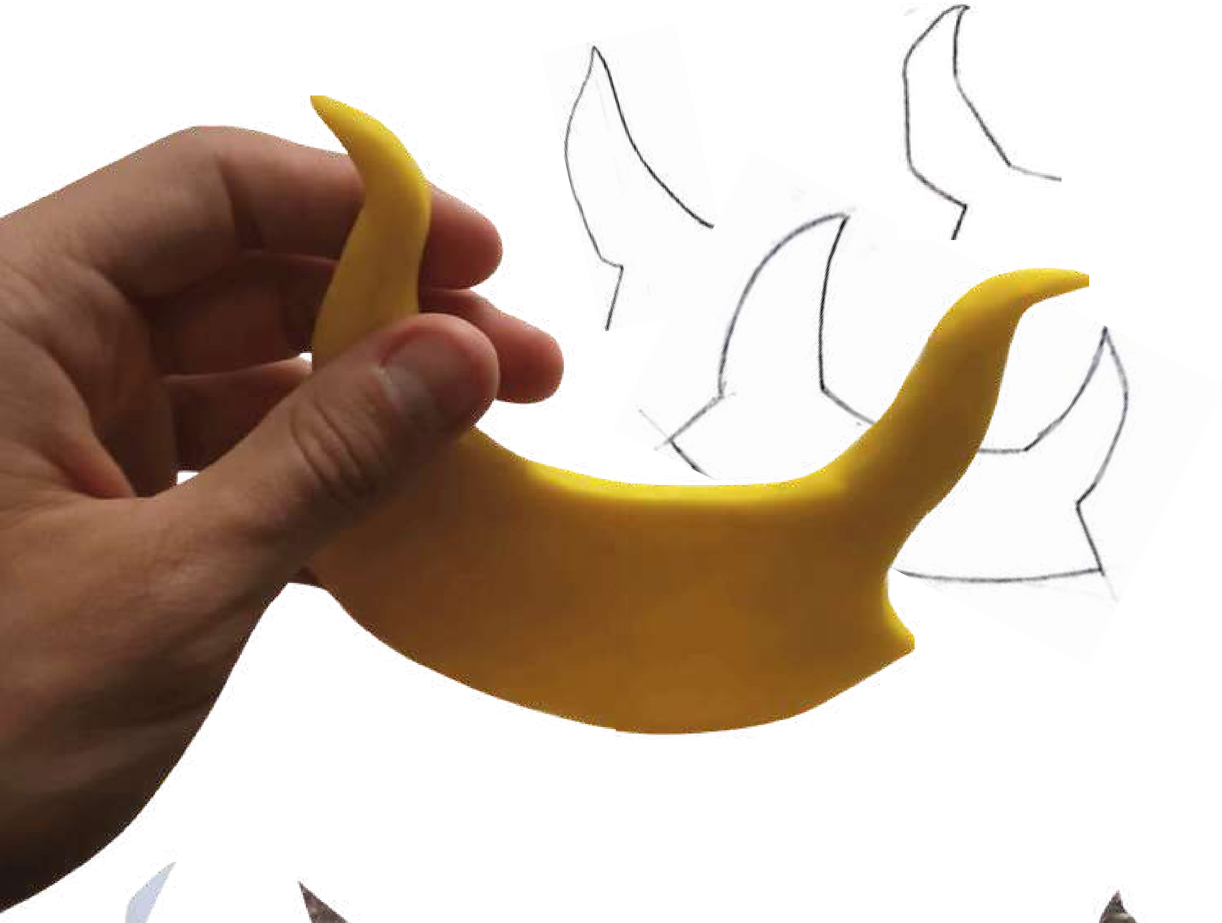
TERRITORY

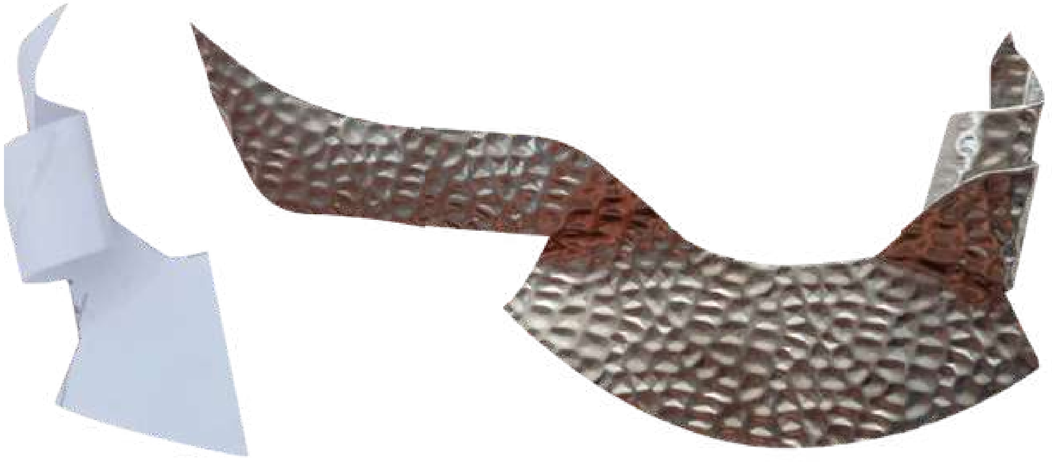
TRADITIONS

STRONG TASTE

COOLNESS

Development





Evocative shape

Handles are the most personalizable part of the object. Through the inspiration of the Boes and Merdules masks, the handles become horns.

Rustic material

The chosen material is hammered aluminum and is strongly linked to the shape of the object, because alluminium sheet is rolled like horns.

Mono

The prototype is monomateric and mono-component, made with just a rolled aluminium sheet. The final object is intended to be made of hammered steel, and be rolled by cold forming, with all the edges chamfered, except the blade edge.

Development

Minimalism

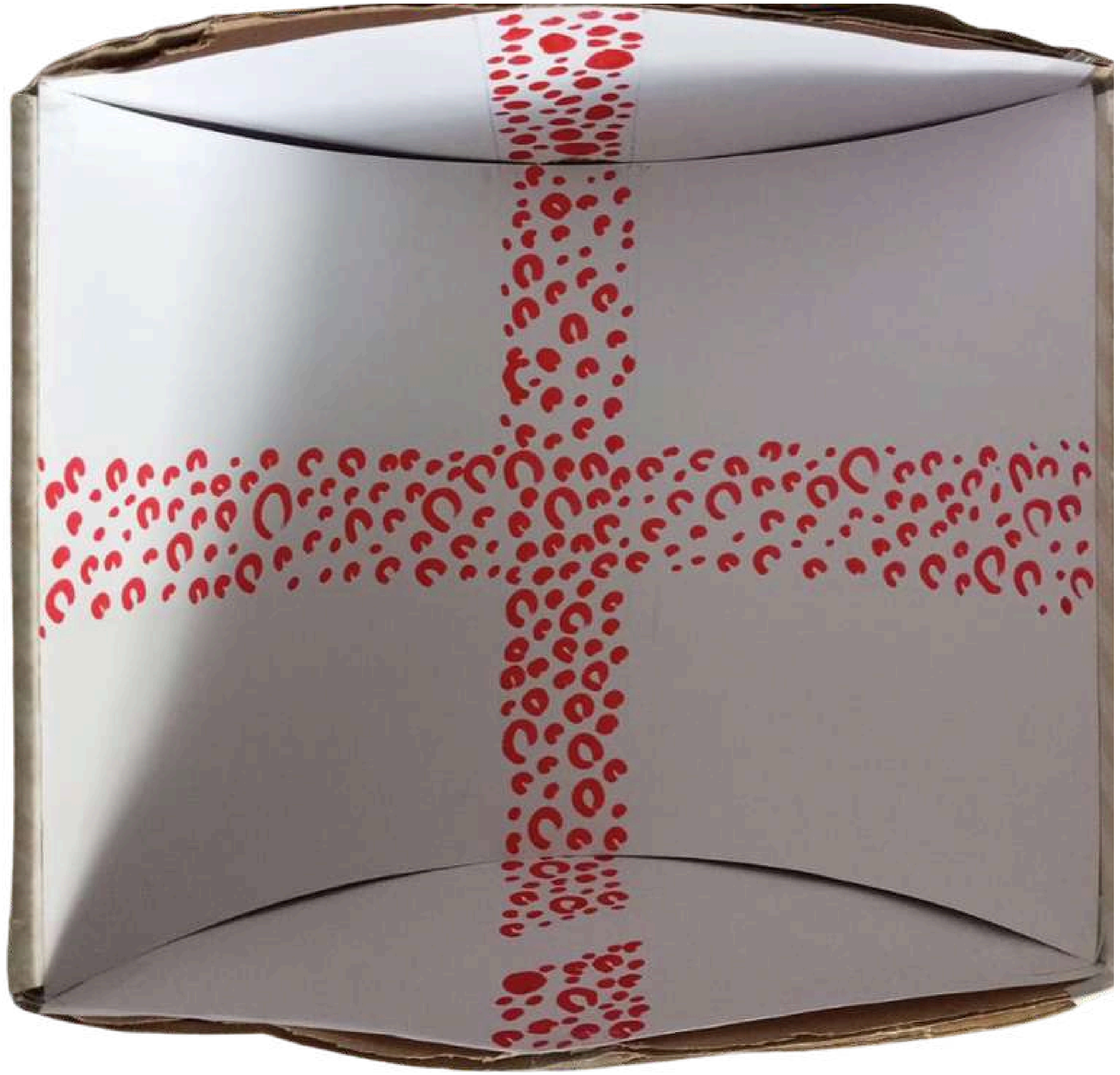
The design of the cube is a simple cross with beer bubble pattern inside. The pictogram comes from the shape of the knife.

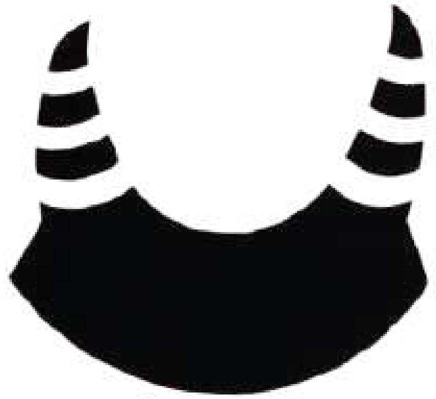
Heritage

The cube is set up like a Greek amphitheater, covering the cubic geometry with the sheet.
The design of the cube is inspired by the flag of Sardinia, just like the logo of Ichnusa.

Consistency

The cut on the bottom strip is linked to the function of the knife, that is inserted in a real cut on the wood cube. This configuration simulates the movement of the object.





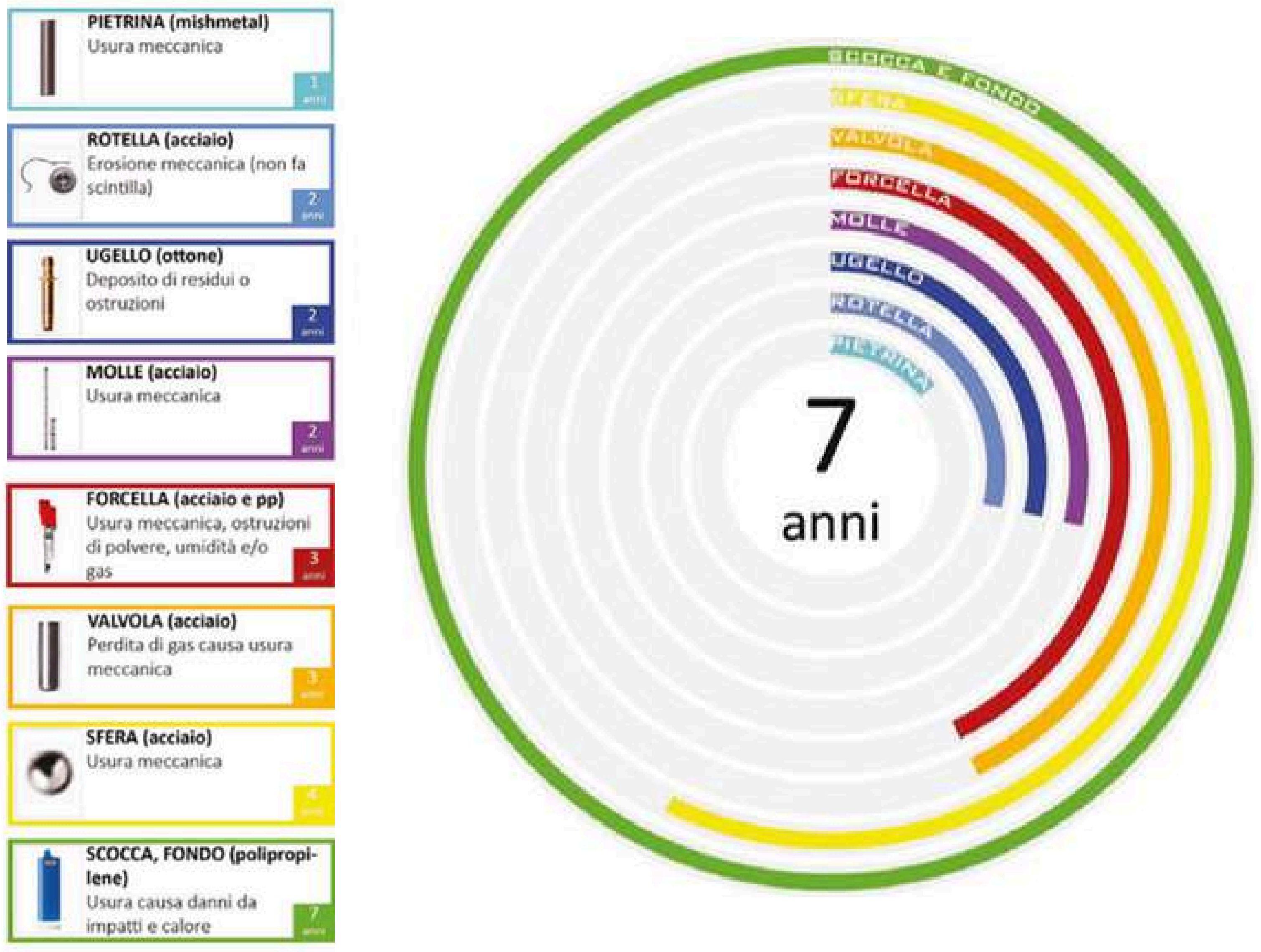
BIC LIGHTER

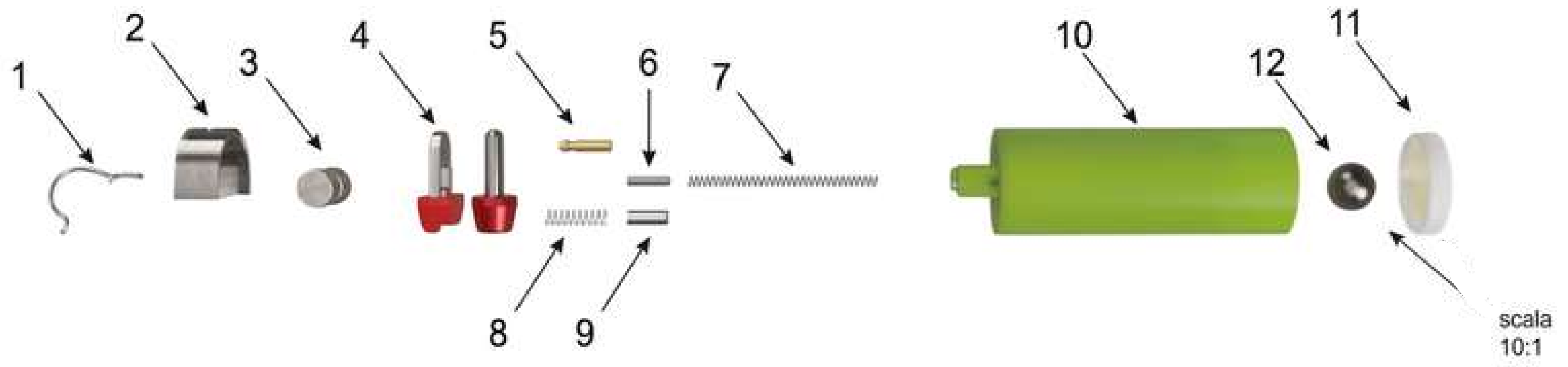
Redesign

This project aims to redesign an object - I chosed the BIC lighter - to improve sustainability, by applying circular economy principles like life cycle analysis and flowchart. the goal is to optimize its complex components and reduce its evironmental impact from production to disposal.

OBJECT ANALYSIS

Lifespan cycle





n	Tot	Component Name	Material	Process	Dimensions (mm)	Weight (g)	CO_2 Eco-cost per Kg (per 1000 products)
1	1	PROTECTION	STEEL	ROLL FORMING	0.30	0.30	1.6
2	1	COVER / LID	STEEL	INJECTION MOLDING/DEEP DRAWING	9.1	9.1	9.1
3	1	FLINT WHEEL	HARDENED STEEL	MIM	9.4	9.4	9.2
4	1	FORK	STEEL AND POLYPROPYLENE (PP)	MIM	4	4	4
5	1	NOZZLE	BRASS	TURNING	0.28	0.28	0.5
6	1	FLINT	MISCHMETAL (IRON-CERIUM ALLOY)	EXTRUSION	0.35	0.35	9.3
7	1	FORK SPRING	STEEL	FORMING	0.22	0.22	1.2
8	1	FLINT SPRING	STEEL	FORMING	0.22	0.22	1.2
9	1	VALVE	STEEL	EXTRUSION	0.12	0.12	0.6
10	1	BODY / CASING	POLYPROPYLENE (PP)	INJECTION MOLDING	9.8	9.8	16
11	1	BALL	CHROME STEEL	INVESTMENT CASTING	r=0.4	0.2	Approx. 0
12	1	BOTTOM / BASE	POLYPROPYLENE (PP)	INJECTION MOLDING	1.03	1.03	1.7

The analysis includes also a flowchart that will change following the redesign of the lighter and a market analysis, that studies the cases of Clipper and Prof.

Development

● → ●●●●

Design for reuse

Short component lifespan, maintenance and disasembly not planned and low attachment to the product

● → ●●●●●

Design for repair

To promote reuse, BIC must allow users to easily replace consumables (gas and flint). Like Clipper, this requires a bottom refill valve and a removable spark wheel assembly for easy access and component replacement.

●● → ●●●●

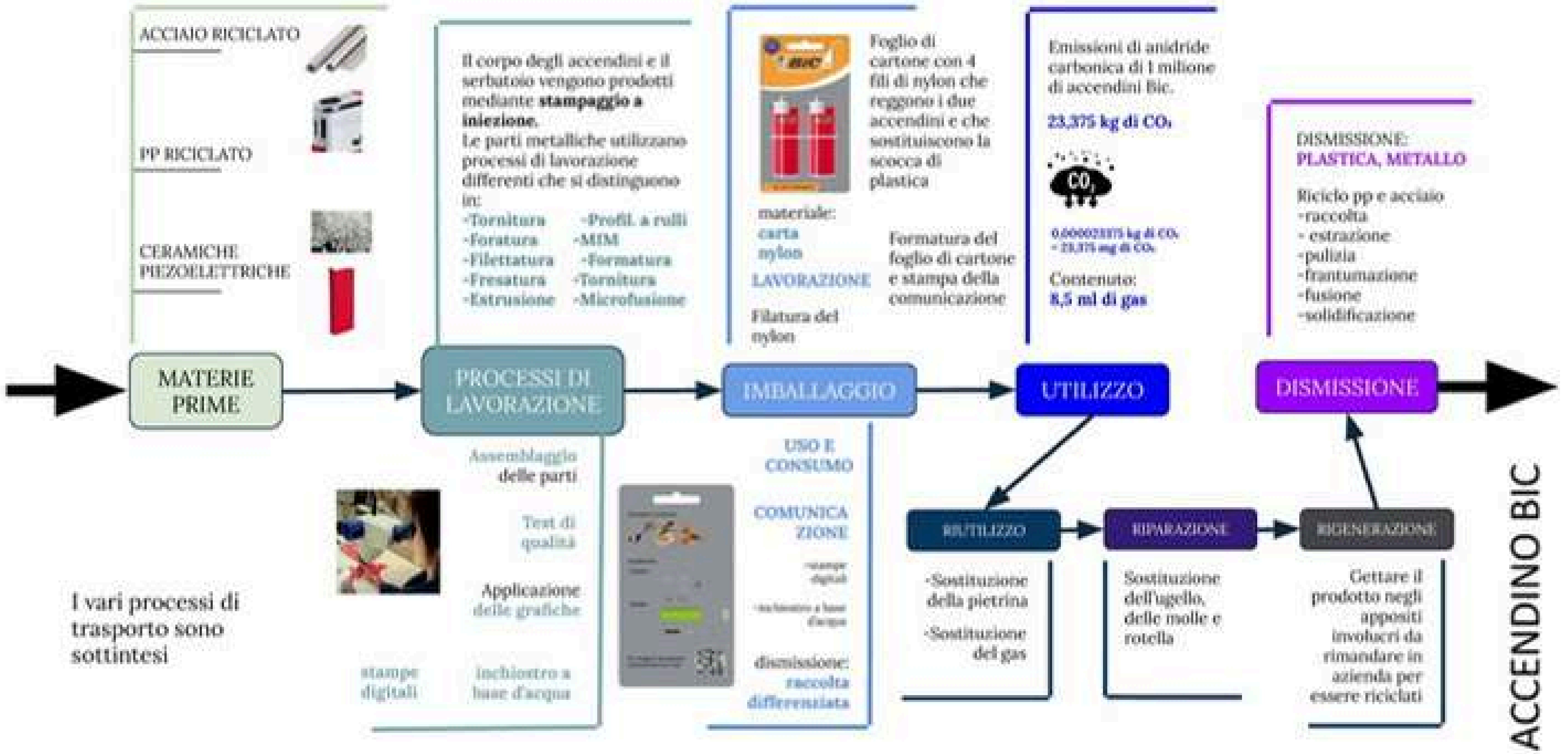
Design for recycle

To improve recyclability, BIC must replace complex material combinations and harmful rare earths with 100% recycled steel, 30% recyclied polypropylene and piezoelectric ceramics.

Building an emotional connection between users and products promotes reuse and repair.
Following the Clipper model, customizing BIC lighters with unique graphics makes them personal and less disposable.
Additionally, using water-based inks ror printing enhances enviromental and economic soustanibility.

Development

Flowchart



The rebuilt lighter is desgined for durability. It features a detachble internal casing that allows for easy flint replacement via a bottom screw. Additionally, a refill valve at the base enables the lighter to be reused multiple times.

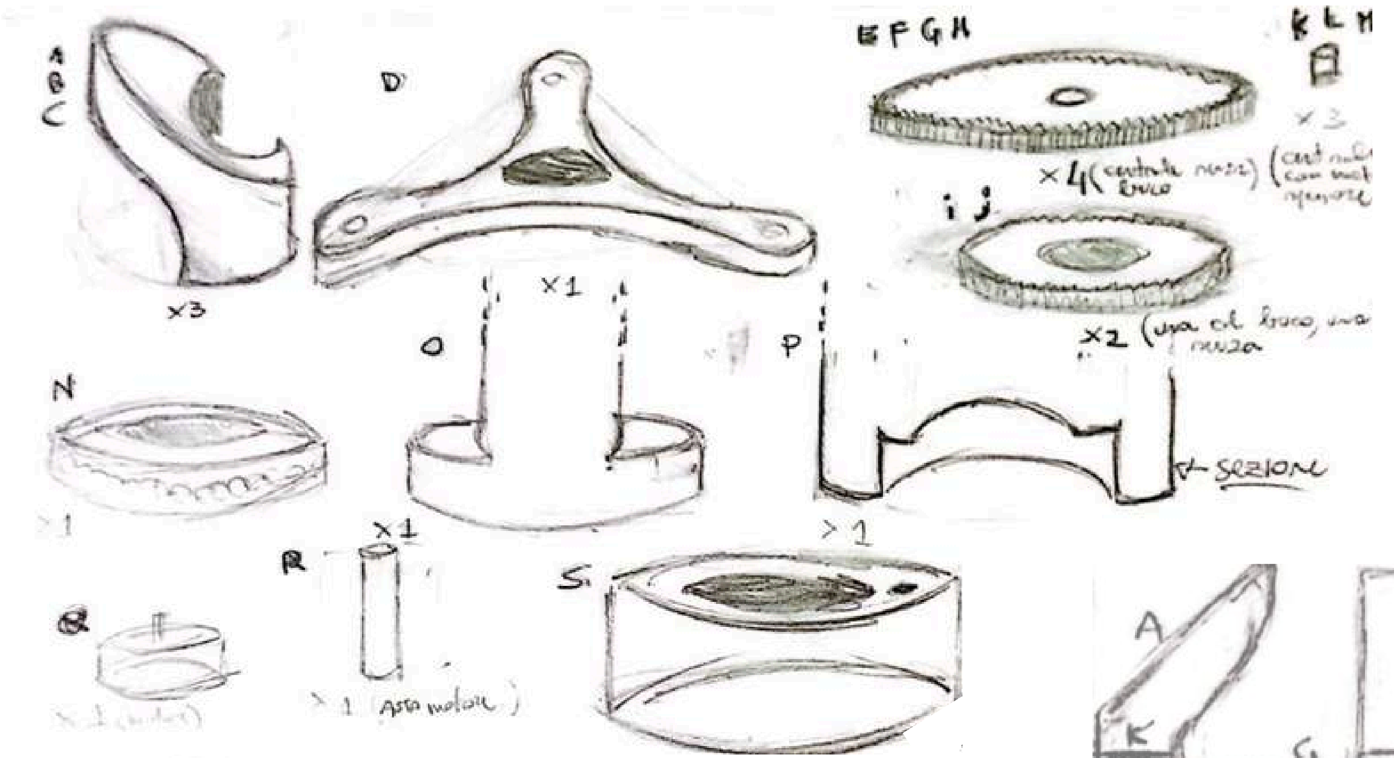
ORBITA

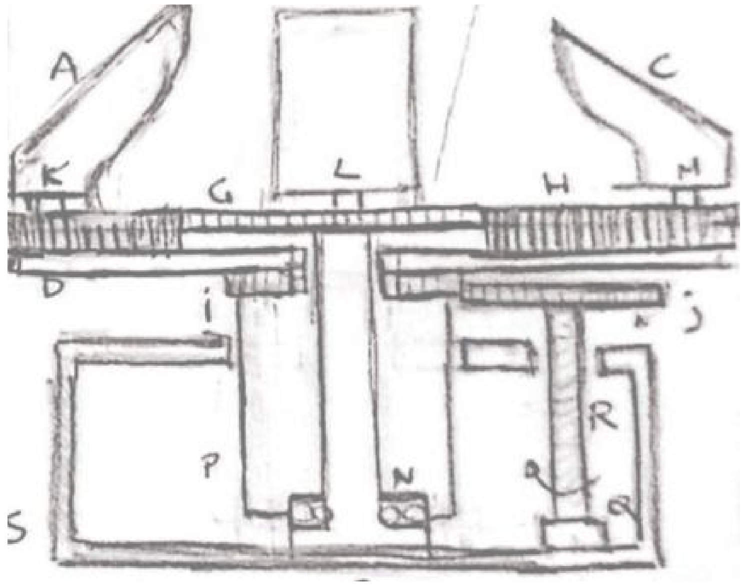
Kinetic sculpture

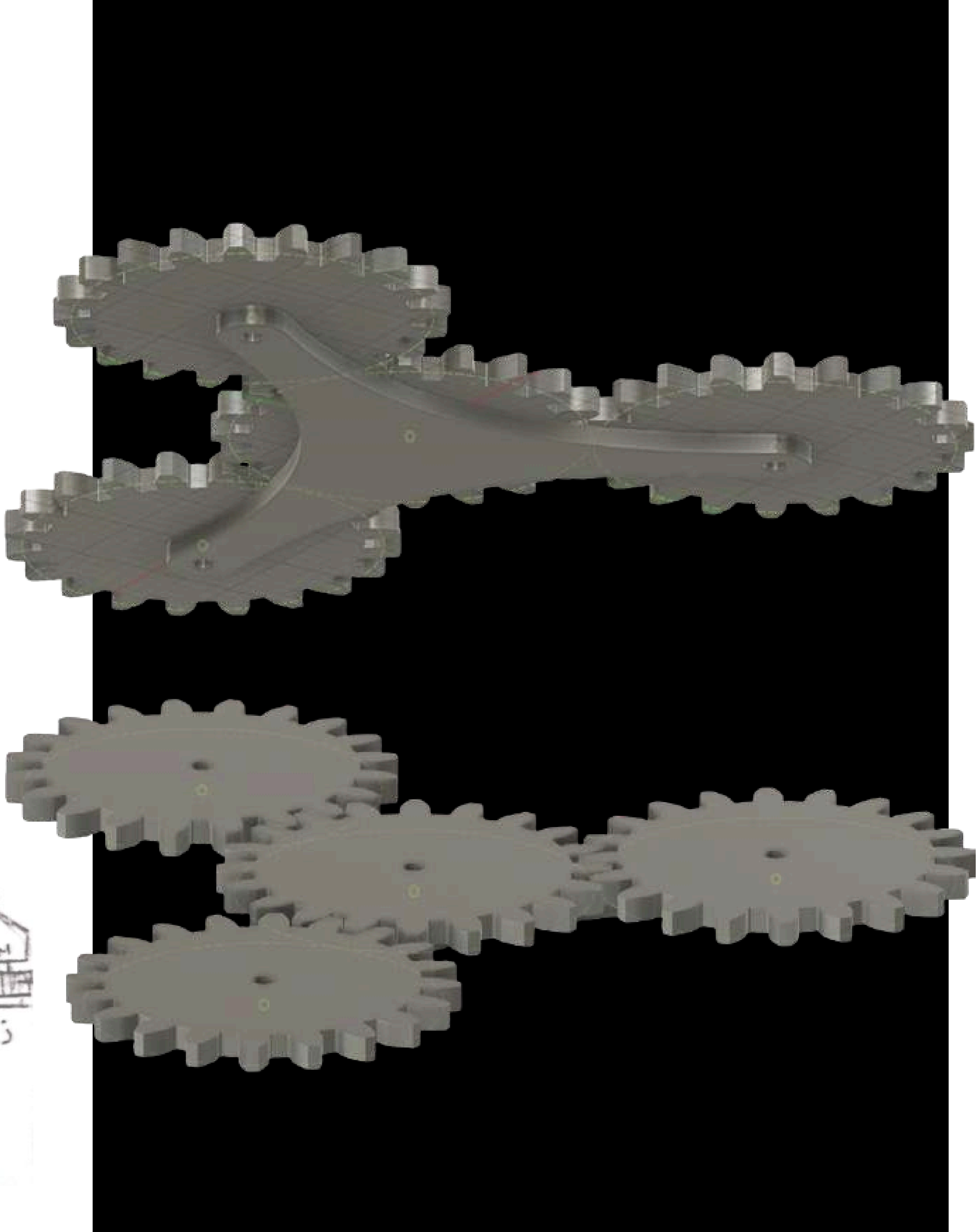
The project involves creating a kinetik sculpture, starting with 3D model and motion simulation in Fusion. The process concludes with a phisical model powered by a motor and gears. The chosen design features three curved elements taht interact through rotation and revolution, moving toward and away from each other.

Mechanical study

To achieve rotation and revolution, a planetary gear system was chosen: a central fixed gear with three outer gears rotating both around it and on their own axes, supported by a rotating three-arm base.
The challenge was to integrate moving components with fixed ones inside a complex gearing mechanism.







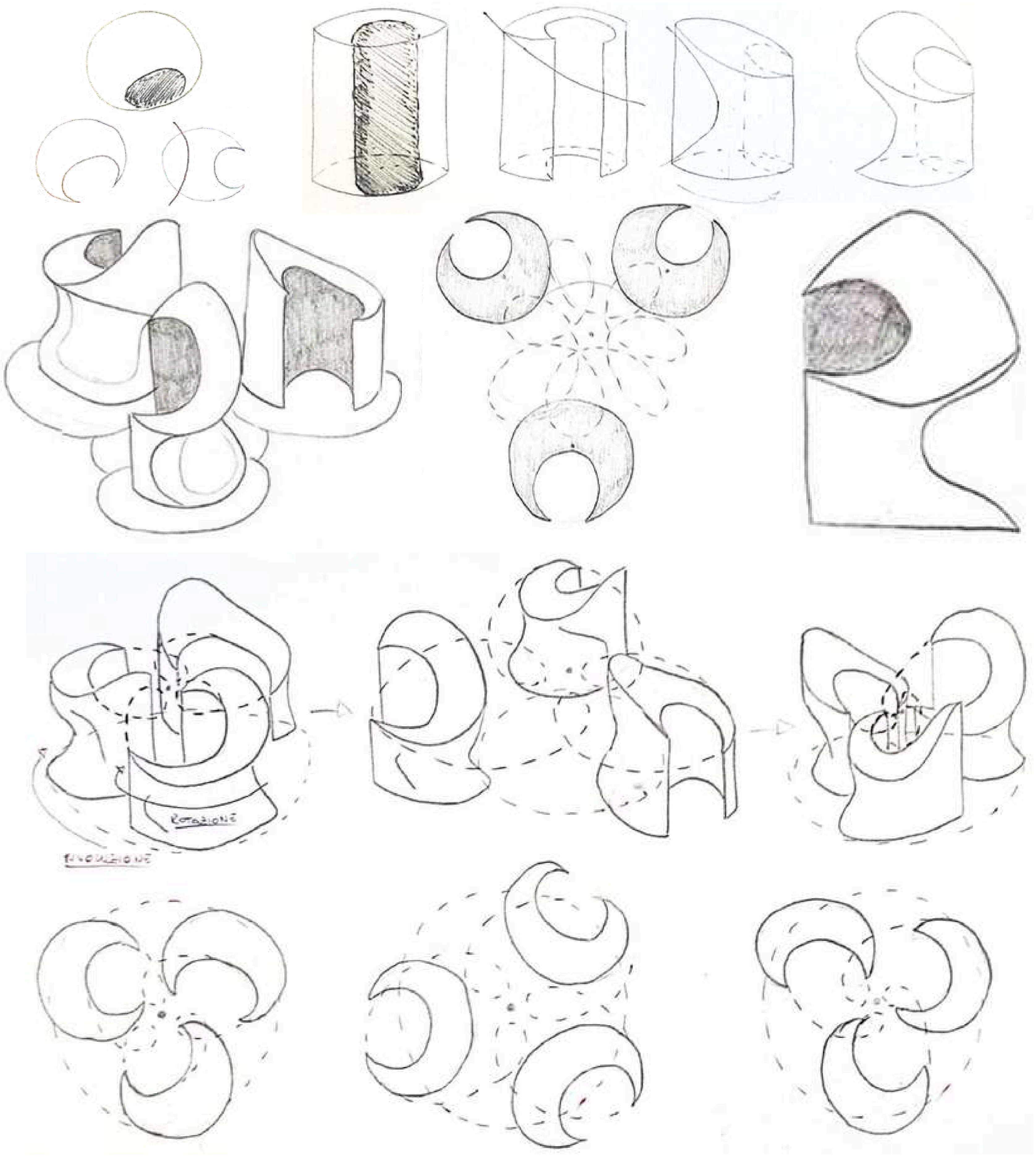
Development

Form sketches

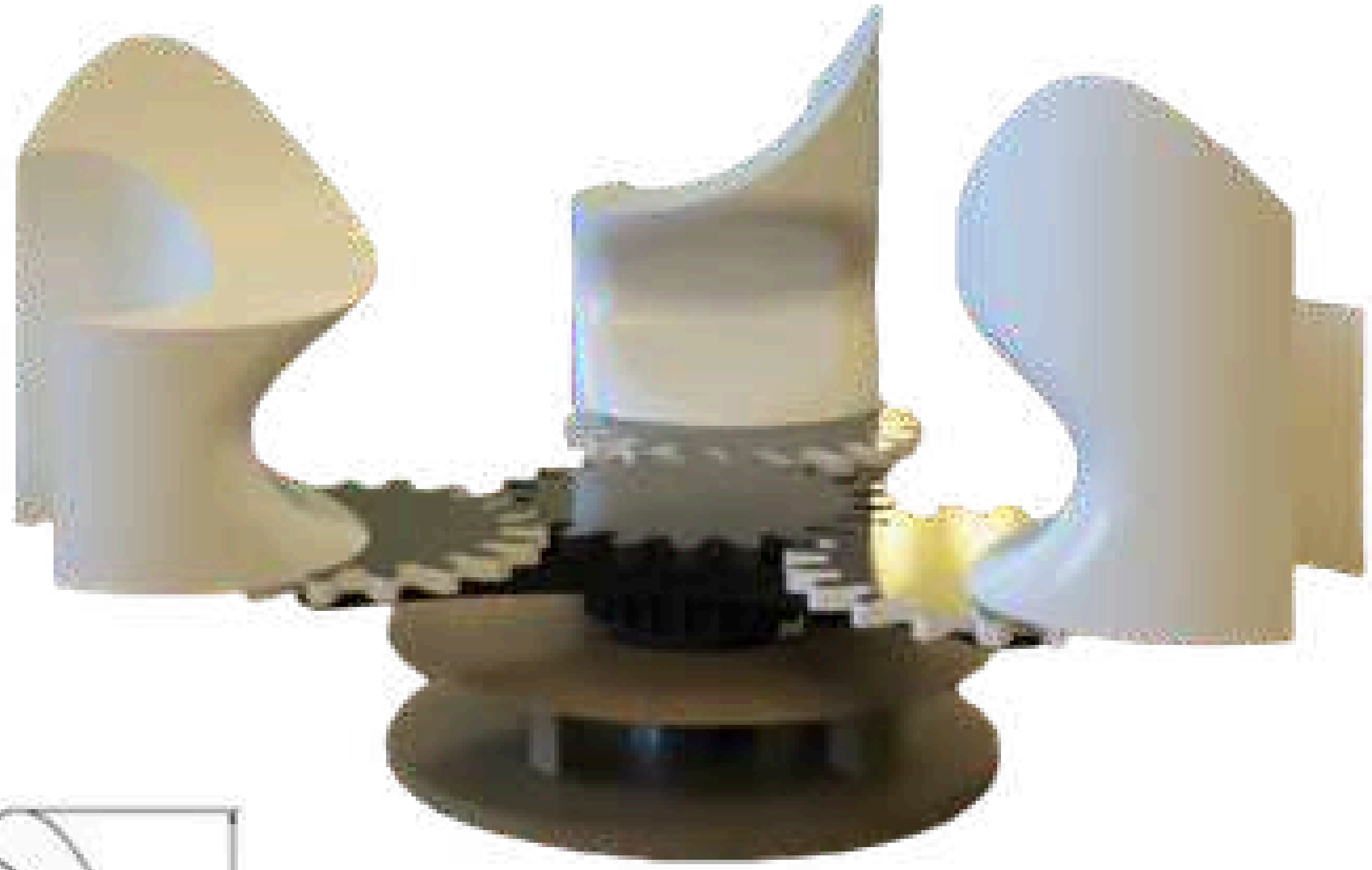
These sketches show the genesis of the form, starting from a cylinder and gradually arriving to a complex organic form.

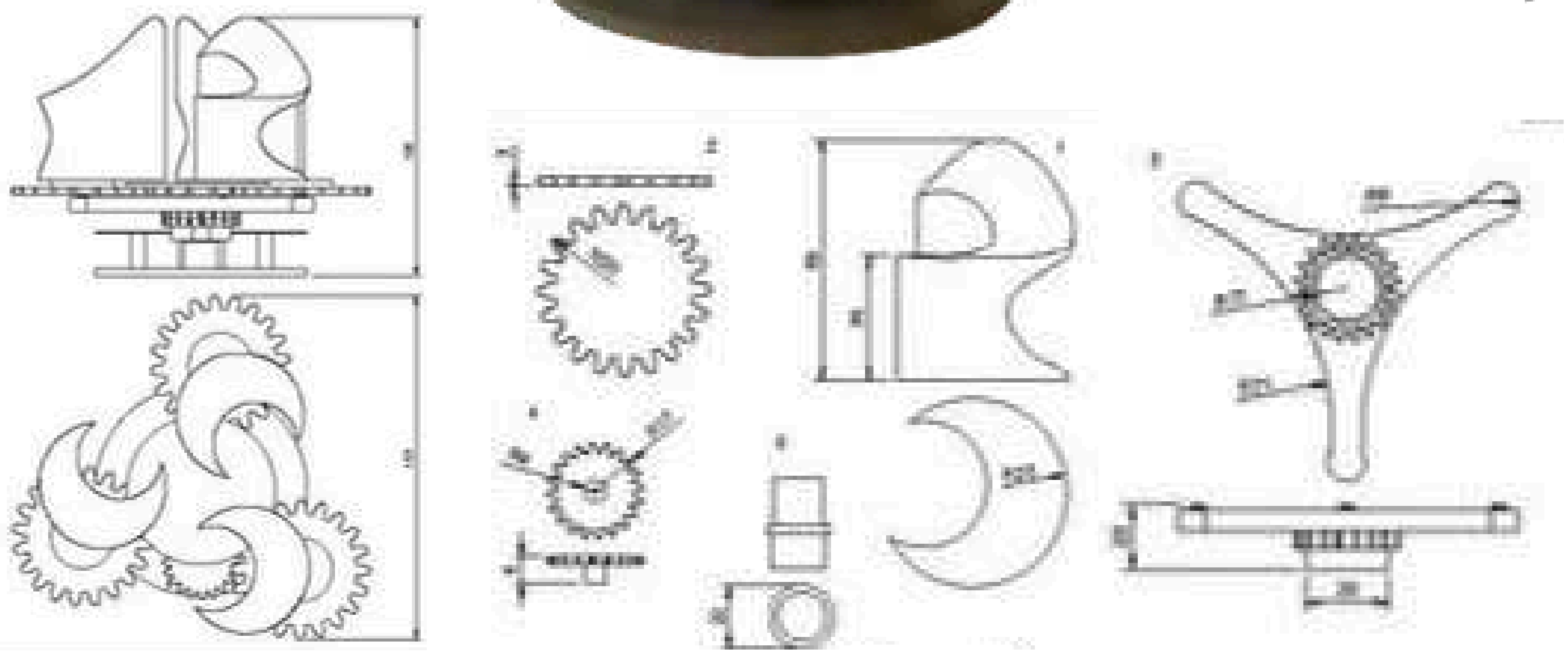
Movement sketches

These sketches show the movement of the sculpture and the way the elements interact to each other, following theirto own elliptical orbit and moving toward and away from the center of the sculpture and from each other.



Development



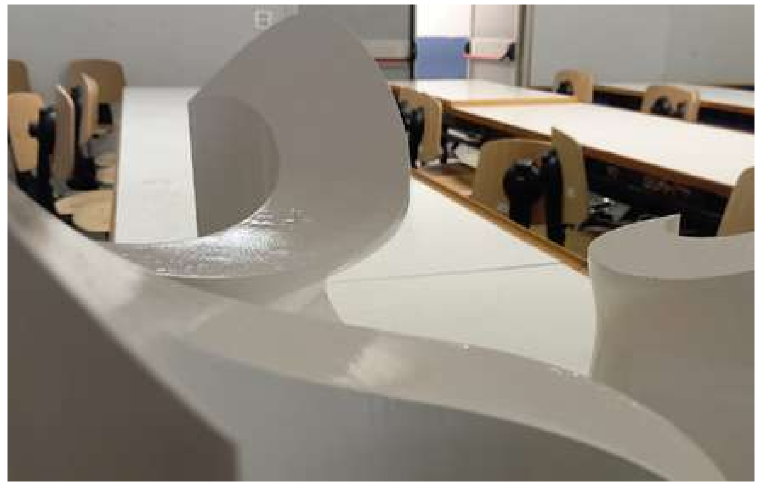


Cad

Development

Prototype

After sourcing motors and bearings, the gears and pillars were 3D printed to ensure the precision that manual crafting couldn't provide. Due to stability issues caused by the weight during rotation, wooden supports and a wooden base were added to reinforce the structure and mantain the correct alignment.



Mr. Hyde

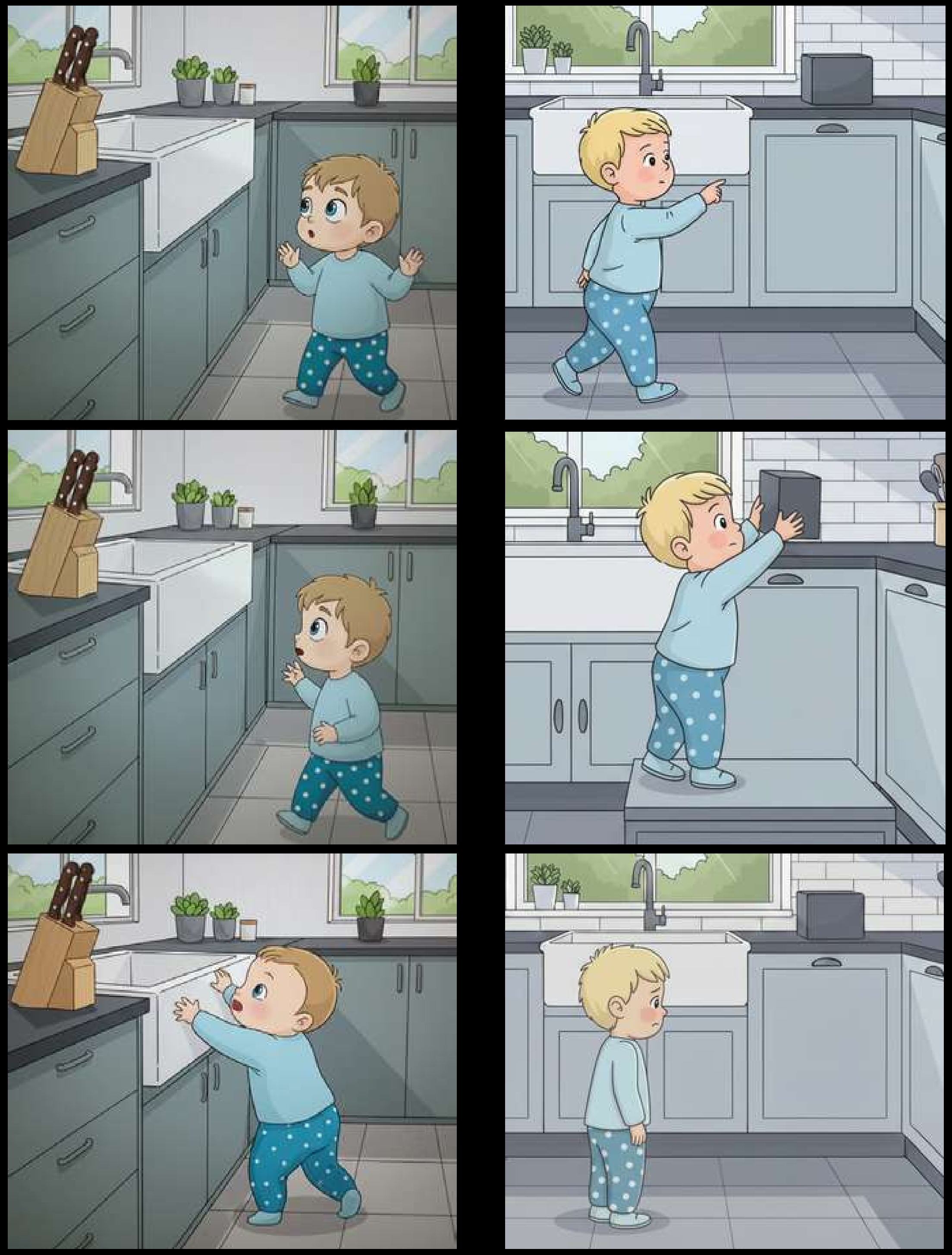
Knife block

The prject involved designing an everyday industrial product focused on the broad concept of containment. Starting from the study of three different objects. Bruno Munari’s “Cubo” was chosen as the primary inspiration. By focusing on its specific”micro function” of hiding rather than just holding, the concept was evolved into the design of a knife block.

Product analysis

As the Cube contains sigarettes to hide them and their smell, the knife block hide knife to make the kitchen safer for the kids.

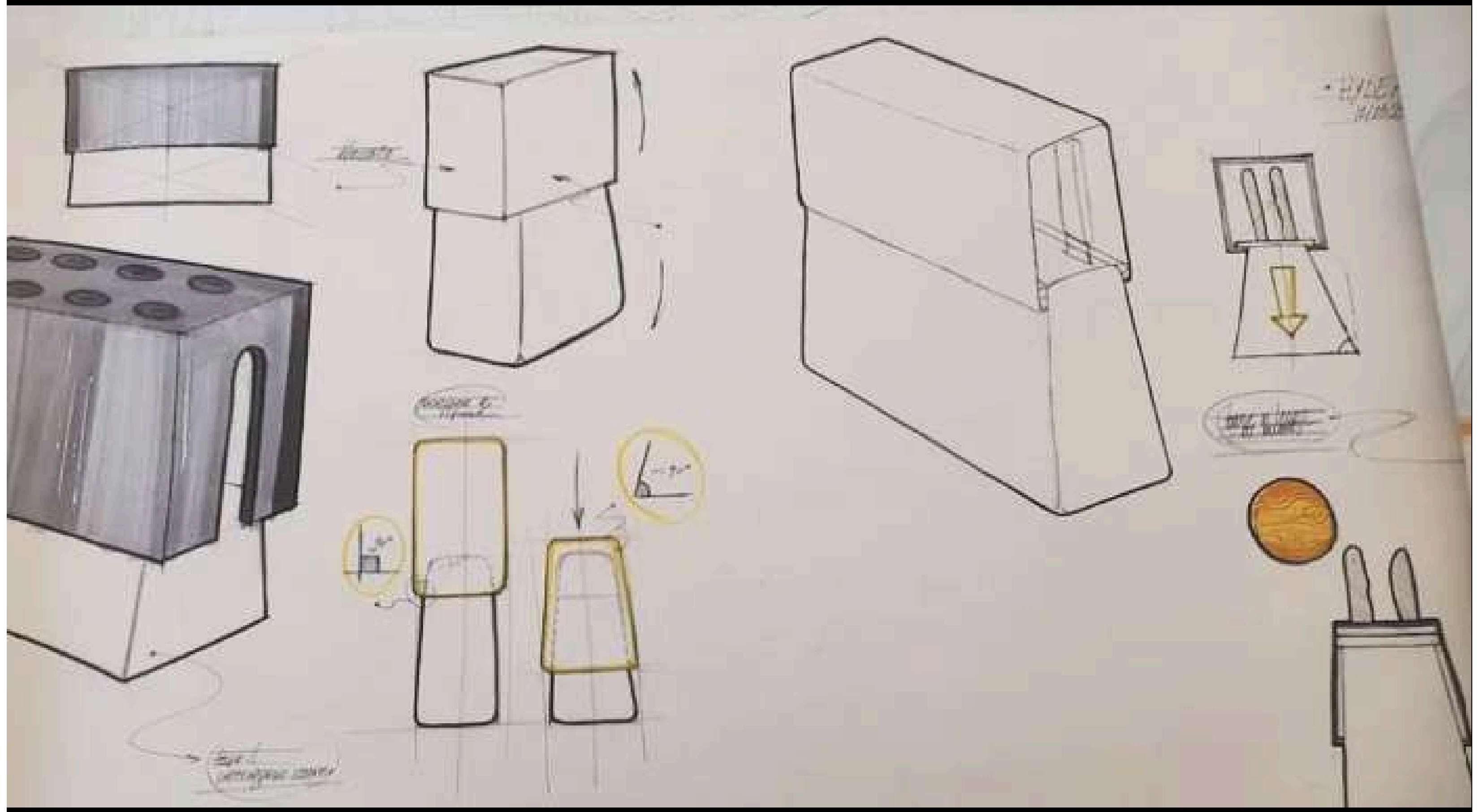
Storyboard



Development

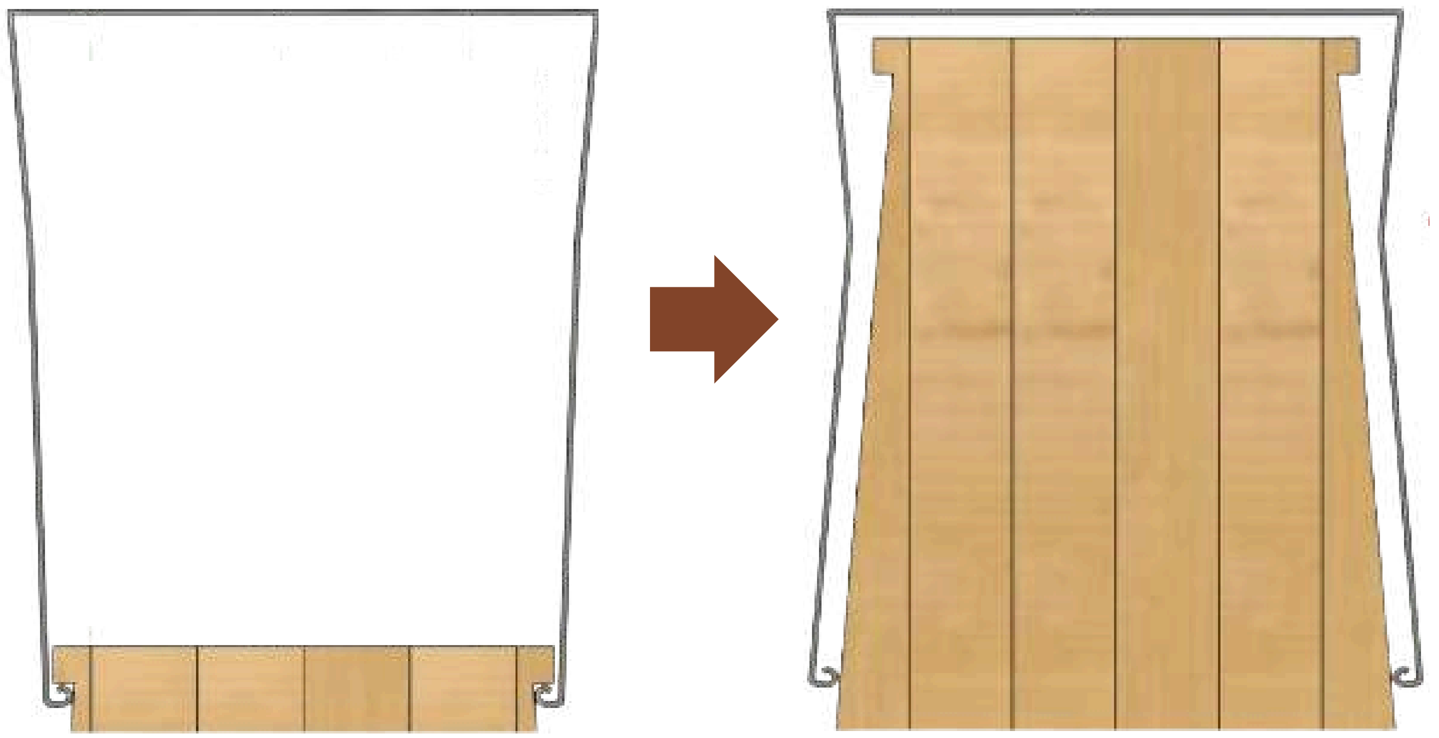
SKETCHES AND PROTOTYPES

Focused on semplicity, this project features a wooden base with sliding metal cover. The mechanism is designed to reveal or conceal handles through a system of integrated holes.



Development

Mechanism

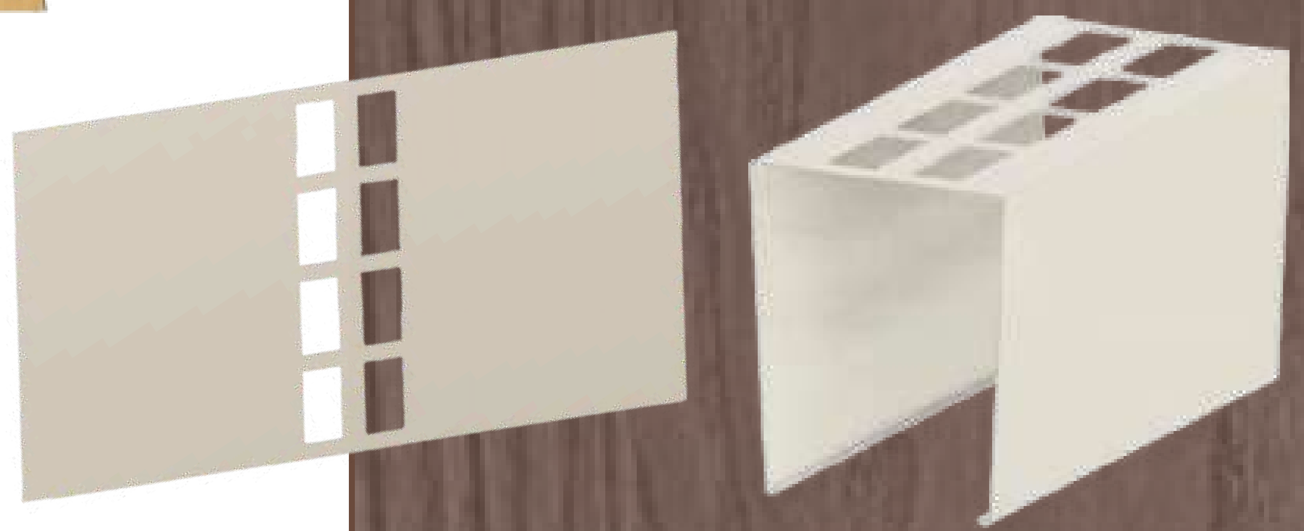


The original position of the sheet metal is at the top, when it covers the handles. When pressure pushes it downwards, it deforms by widening. sliding down and uncovering the knives. The movement of the sheet is supported by two steel bars that slide up and down on the block.

Production

Ash wood

- laser cutting on the panel shapes
- CNC milling
- laser engraving for text



EN AW Aluminium

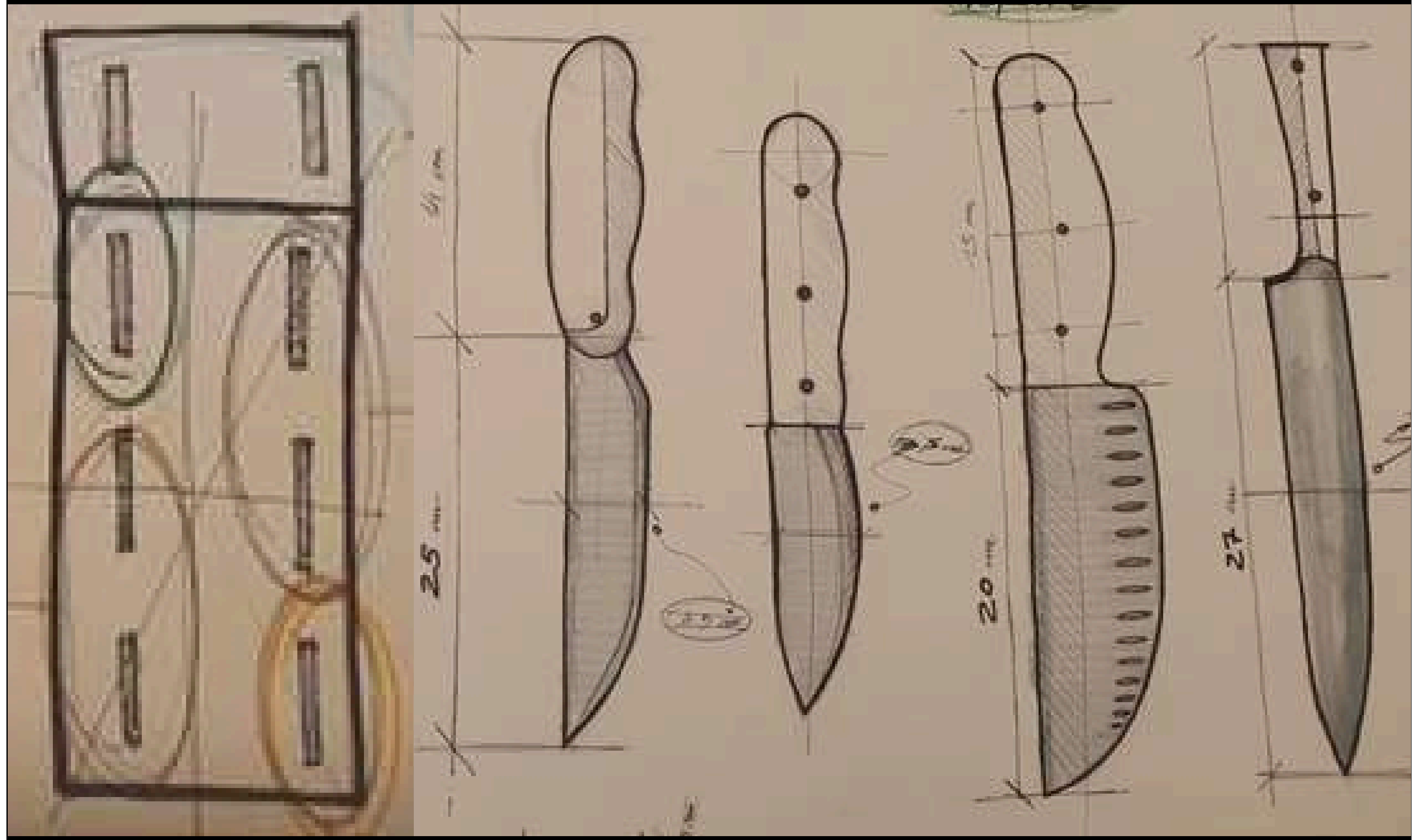
- saw cutting for the sheet metal portion
- laser cutting for holes
- press brake for corner bending

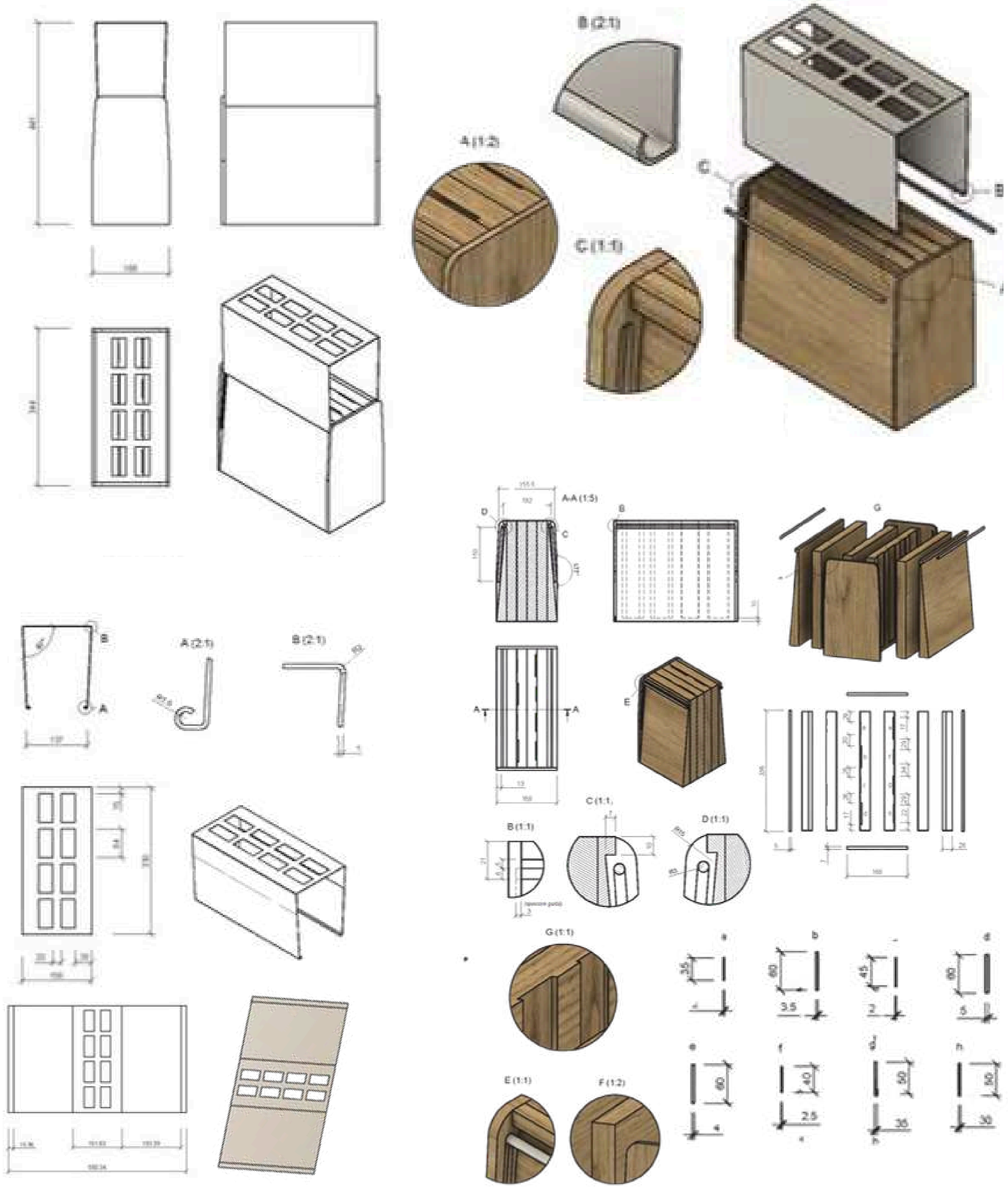
Development

Dimentions

Td

The study ff each type of knife, to size the block and the 8 different holes.

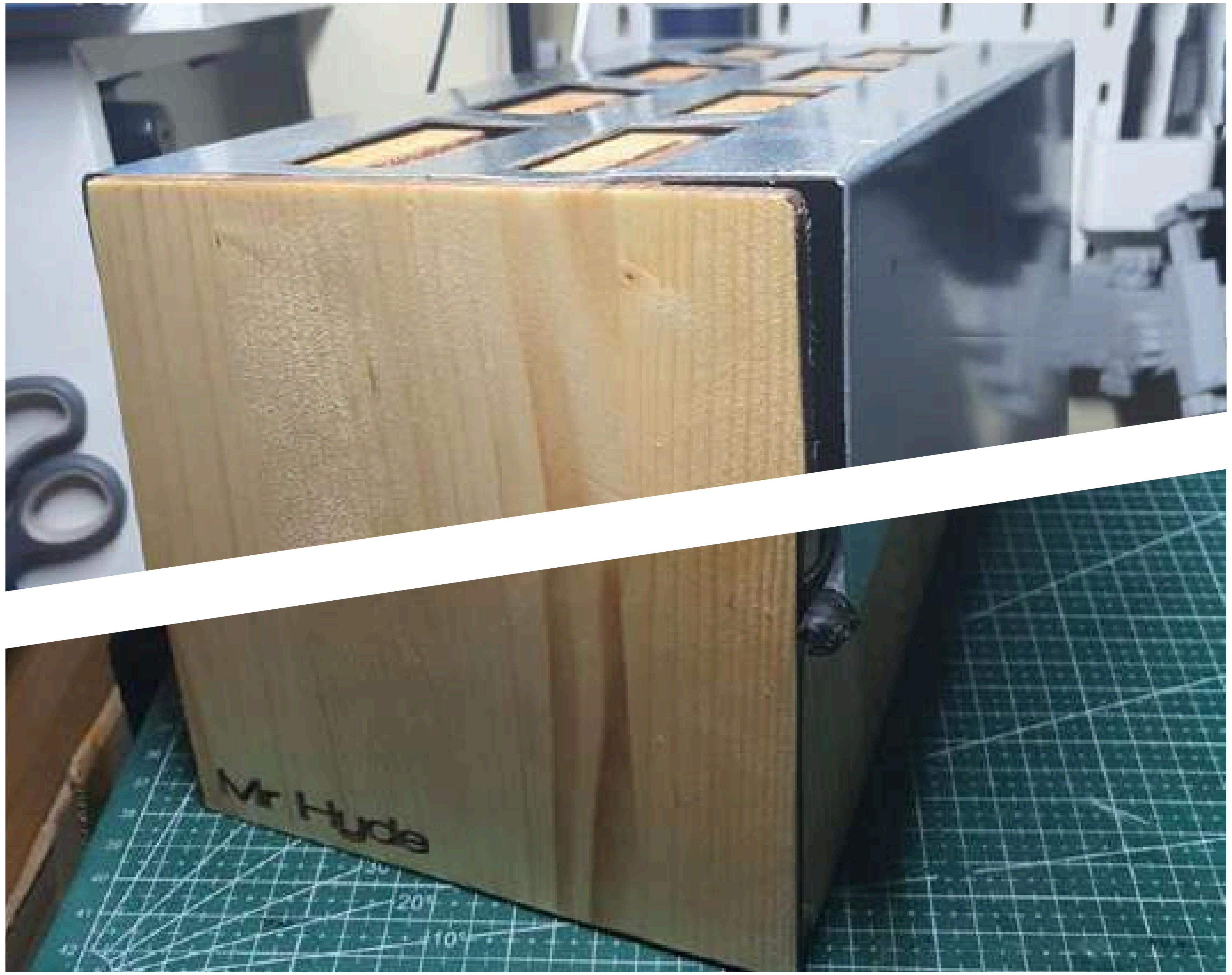


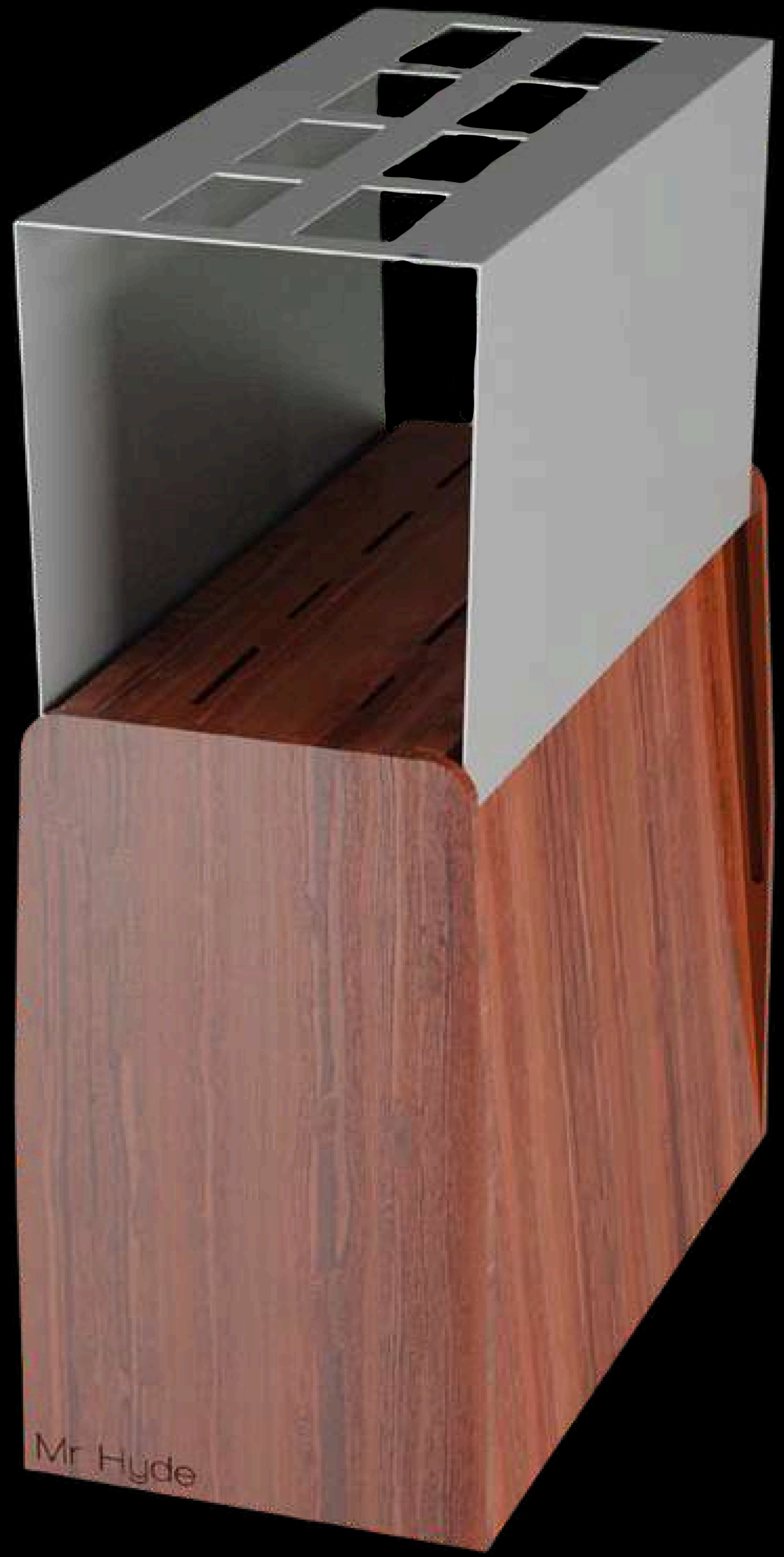


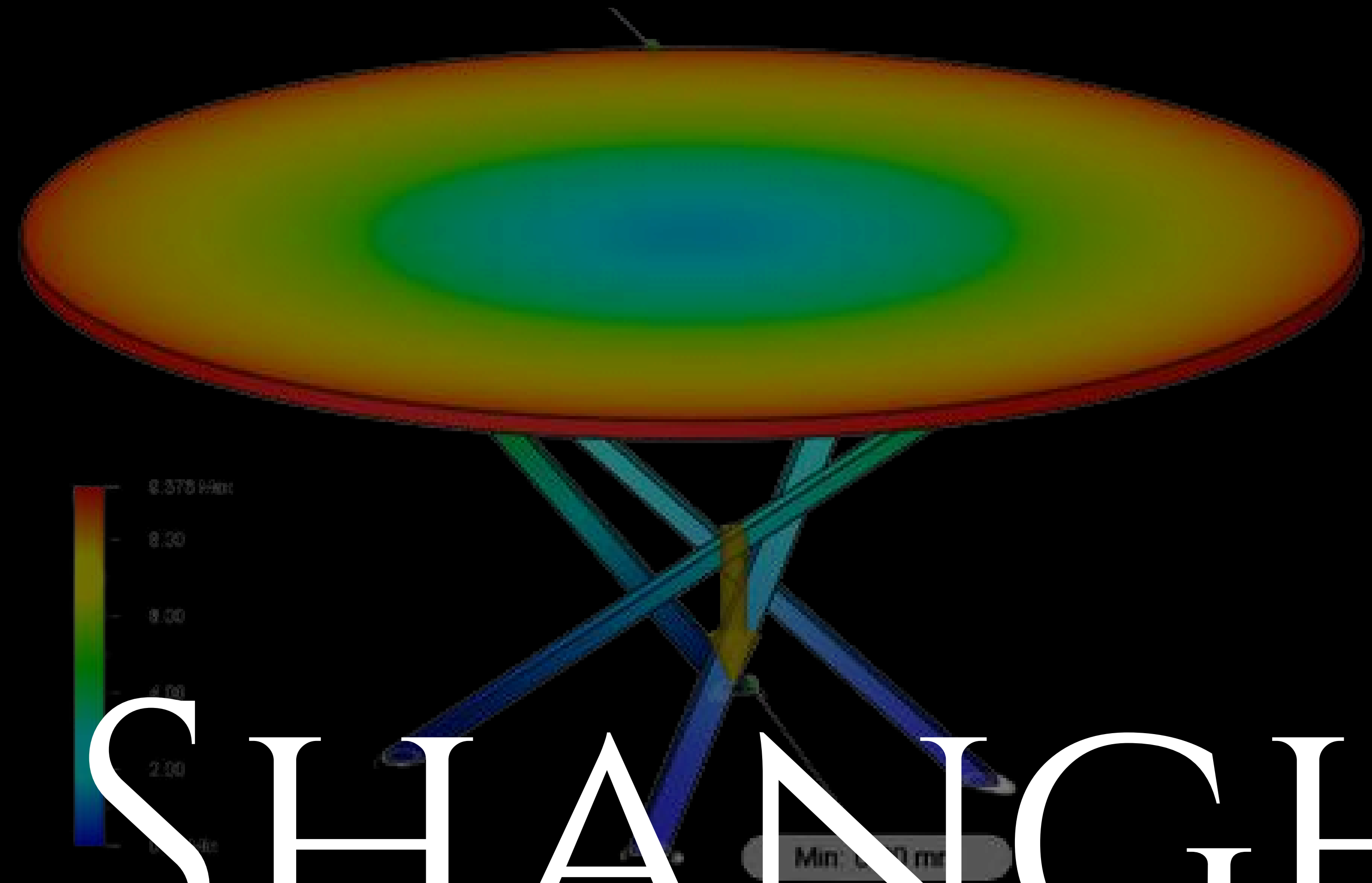
Development

Protoyipe

Rendering







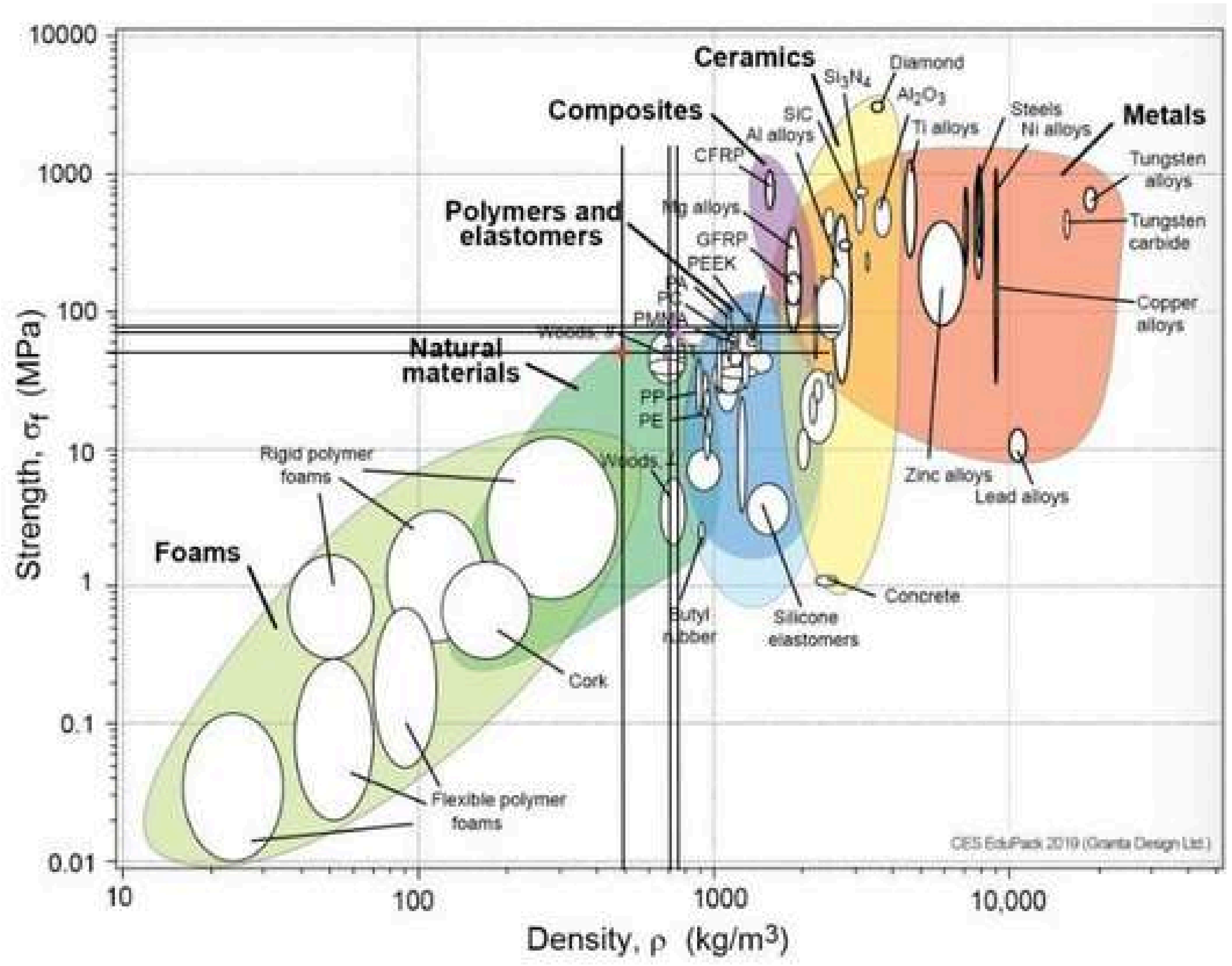
Shanghai

Dining table

The course includes the structural design, of an object with a complex morphology (we chosed this type of table), in order to evaluate stresses and deformations based on the analysis of usage scenarios and mechanical requirements. Through the results of simulations carried out on Fision 360, continuous optimizations have led to defining the optimal combination of shapes, materials, and dimensions.

Materials analysis

Ashby diagram



1.METALLI

Materiale	Densità (kg/m³)	Resistenza a s. (MPa)	Rigidezza (GPa)
Alluminio EN AW-5083	2660	125	70
Acciaio AISI 304	8000	205	193
Acciaio AISI 316	7980	205	193

2.LEGNI

Materiale	Densità (kg/m³)	Resistenza a s. (MPa)	Rigidezza (GPa)
Pino rosso	545	50	11,24
Quercia bianca	755	70	12,15
Faggio europeo	710	80	14,31

3.POLIMERI

Materiale	Densità (kg/m³)	Resistenza a s. (MPa)	Rigidezza (GPa)
Cossa Polimeri ESTACARB 0501 Policarbonato	1200	60	2,52
Nylon GF Caricato con fibra di vetro	1490	26	3,10
ABS ABS834G10L Rinforzato con Vetro	1090	20	2,50

The white oak has been chosen, beacuse it has a good average of features.

Development

MECH. REQUIREMENTS

Data from tables of similar sizes were examined, in order to decide the mechanical requirments of our table.

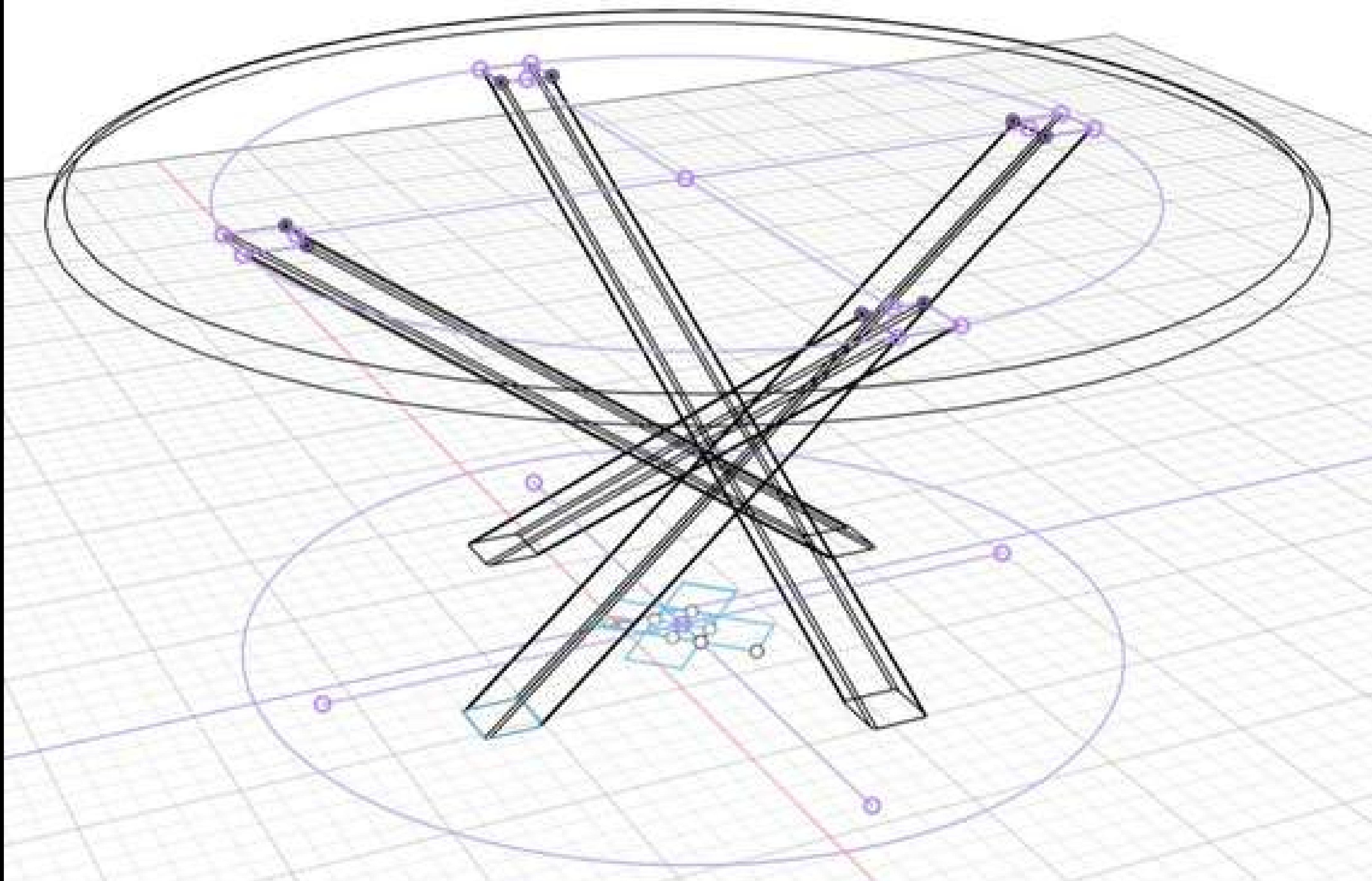
Weight: 50 - 70 kg

Maximum load 1: 1.5 - 2 kN

Maximum load 2: 3 - 3.5 kN

Deformation: 10-15 mm

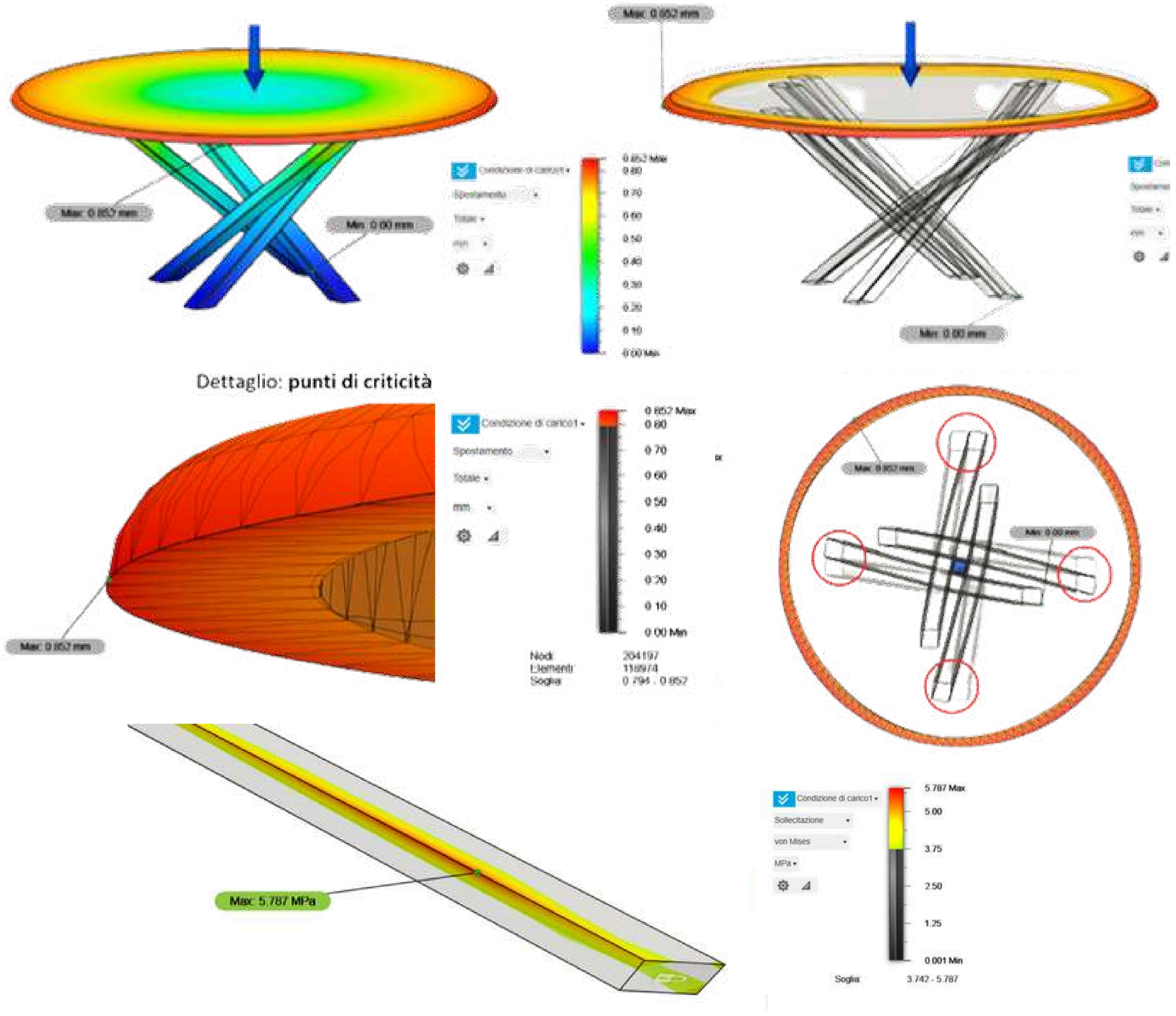
3D MODELING

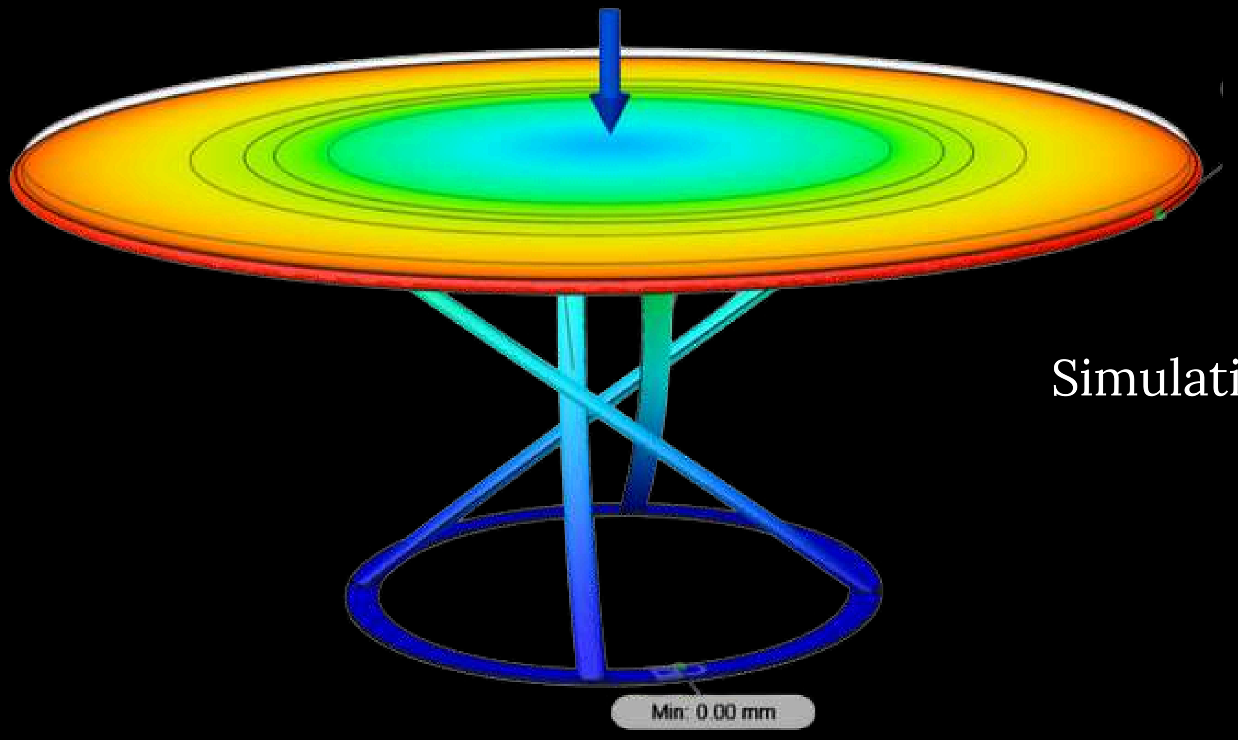


Development

5 Simulations

Simulation 1





Simulation 5

$$Q_{max} = \frac{125\ \text{MPa} - 17\ \text{MPa}}{31\ \text{MPa}} = 3.5\ \text{kN}$$

Max: 14.4 mm

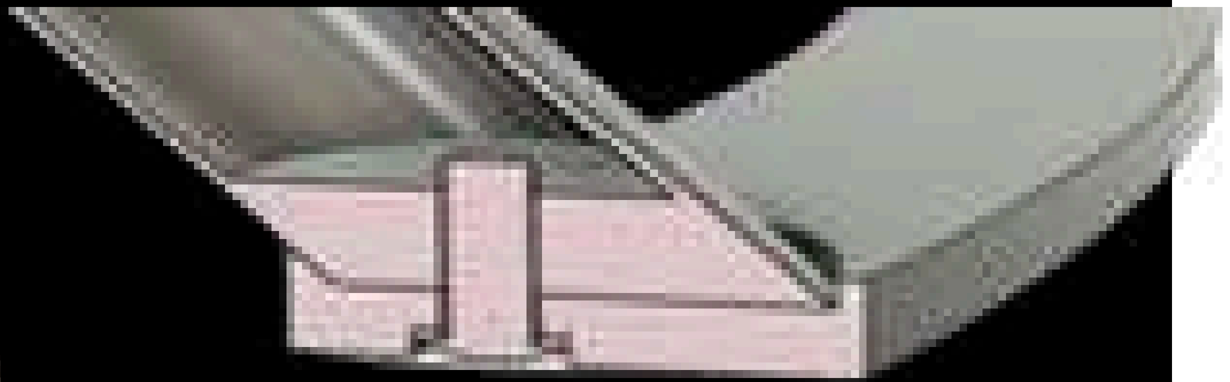
Since it was decided to use a wooden top and metal legs, two separate simulations were conducted. However, only the simulation for the legs was taken into account, as they are the part of the table subjected to the ground reaction force and given that the tabletop is extremely resistant to compression.

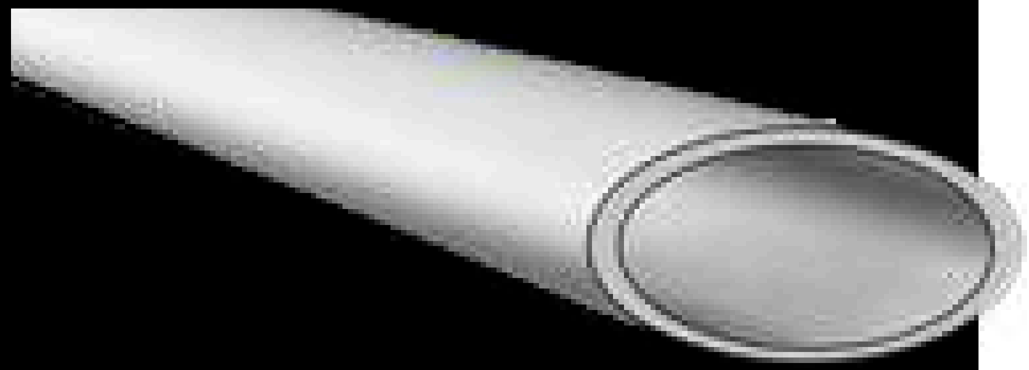
Development

5 OPTIMIZATIONS

Development

Assembly





Trivio

A travel planning app

The brief required designing an app, following UX/UI rinciples and tools while also incorporating strategic branding aspects. The choice was made to design an app that allows users to plan a trip.

ANALYSIS

USER PERSONAS

IRIS MANCINELLI
La ragazza

LE SUE ABITUDINI:
Giulia viaggia per piacere, tre o quattro volte all'anno, di solito d'estate, con le sue amiche. Non ha quasi mai una meta precisa in mente, ma di solito valuta i posti sulla base della comodità, badando poco al prezzo.
Di solito viaggia in aereo e alloggia in hotel o case prenotate su Airbnb.
E' abbastanza pigra e non ha voglia di organizzare nulla, a meno che non sia strettamente necessario. L'unica cosa che le interessa è che le mete siano instagrammabili e abbiano buone discoteche.

ETA': 18 anni
PROFESSIONE: studentessa

" Non voglio sbatti ad organizzare viaggi, sia chiaro...qualcuno pensi a spostamenti e prenotazioni, grazie!

FRUSTRAZIONI:
- Odia le camere piccole e in particolare i bagni scomodi
- Non le va di organizzare, ma allo stesso tempo è molto esigente
- Macchina e treno sono scomodi

BISOGNI:
- Le piace dare indicazioni per cosa vuole in una vacanza
- Vuole andare in posti adatti a ragazzi/e della sua età
- Vuole itinerari pronti e con i mezzi

ROBERTO PIERSANTE
Il fuorisede

LE SUE ABITUDINI:
Tommaso viaggia per piacere, cinque o sei volte l'anno, con i suoi amici o la sua fidanzata. E' disposto a rinunciare alla comodità pur di non spendere troppo. Nei suoi viaggi abbina divertimento all'aspetto culturale e naturalistico. Preferisce quindi mete che gli permettano di soddisfare tali interessi.
E' aperto a tutti i mezzi di trasporto e ad alloggi modesti. E' molto attento al rapporto qualità/prezzo, cosa che spesso gli rende l'organizzazione di una vacanza troppo impegnativa, considerando gli impegni universitari e lavorativi.

ETA': 25 anni
PROFESSIONE: studente

" Con tutte le spese che ho non posso vendere un rene per le vacanze! Se sono organizzate bene, ti diverti e spendi pure poco.

FRUSTRAZIONI:
- Detesta quando va fuoribudget
- A volte non trova mete adatte ai suoi interessi e ai suoi bisogni
- Capita che delle mete deludano le aspettative che si era fatto

BISOGNI:
- Ha bisogno di qualcuno che lo aiuti ad organizzare.
- Vuole andare in posti con vita notturna, musei, paesaggi
- Confronta tutti i tipi di mezzi e alloggi

ANDREA LUCARINI
L'uomo indaffarato

LE SUE ABITUDINI:
Andrea viaggia sia per lavoro, quando è la sua azienda ad organizzare i viaggi, sia durante le vacanze; ed è qui che sorgono i problemi. Non riesce a trovare il tempo per organizzare vacanze che coniughino i suoi bisogni con quelli della sua famiglia, finendo per doversi affidare ad agenzie di viaggio accontentandosi delle mete e dei viaggi propinati. Lui infatti è elastico e si adatta facilmente, ma non sua moglie e i suoiSpesso, non riuscendo a trovare nulla che metta d'accordo tutti, ripiega per viaggi di piccola entità.

ETA': 53 anni
PROFESSIONE: commercialista

" Mia moglie vuole che sia io a pensare alle vacanze, ma poi si lamenta per qualunque cosa. Come se avessi tempo per 'ste cose!

FRUSTRAZIONI:
- Detesta deludere le aspettative della sua famiglia
- Non ha il tempo per accontentarli
- Non è bravo con la tecnologia e comprendere app e siti web

BISOGNI:
- Ama posti particolari ed esotici
- Gli serve un buon seivizio clienti
- Essendo preciso, ha bisogno di un sistema di prenotazioni ed itinerari altrettanto organizzato

COMPETITOR

Nomads

Rich info, efficient advanced filtering

One-time payment, niche audience

Tripomatric

Strong community

Poor personalization

Evaneos

High quality, curated experience and UX, premium perception

Non-scalable business (consulting)

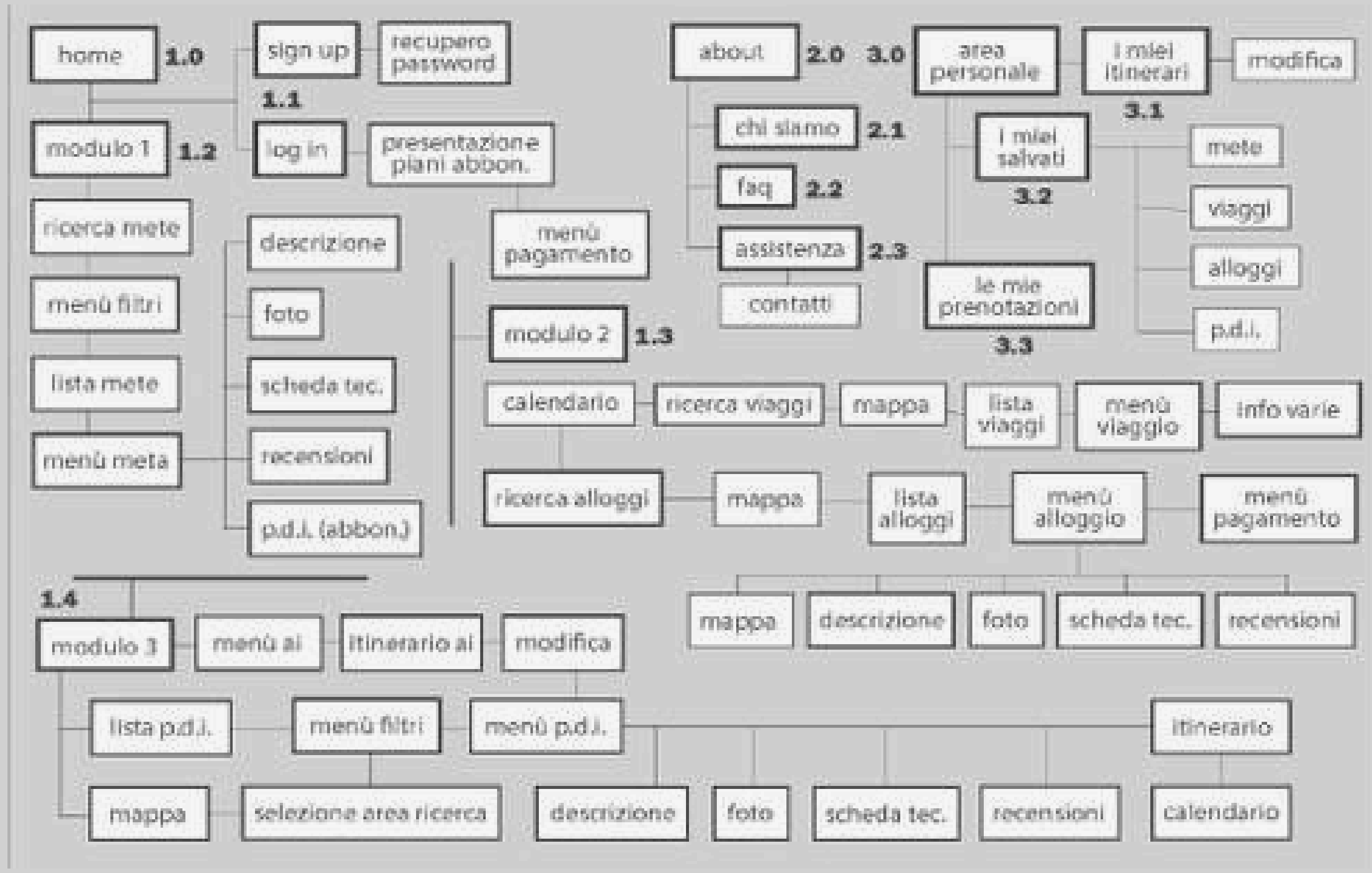
Ux development

Customer journey

	RICERCA META	PRENOTAZIONI	RICERCA ATTIVITA'	SCELTA ITINERARI
AZIONE	Ricerca online logistica Ricerca online opportunità Brainstorming	Tramite siti online o contatti telefonici	Brainstorming Ricerche online	stesura degli itinerari manuale o tramite app o siti
PENSIERO	E' difficile organizzare viaggi: bisogna gestire i bisogni di tutti, cercare i posti e le opportunità che o rono e contemporaneamente controllare voli e alloggi, poi con 100 schede aperte...	Finalmente posso prenotare. Speriamo vada tutto bene...	Ahh, divertente cercare le cose da fare, peccato che ci siano vari siti per ogni cosa e siano consigliate sempre le stesse cose. Spesso non danno tutte le info e le devo cercare io!	Finalmente è tutto pronto e organizzato. Sarebbe più comodo se ci fosse un modo più automatico ed elastico per fare gli itinerari, anche nel caso dovessi modi care qualcosa.
SENTIMENTO	Entusiasmo Frustrazione da indecisione Stress da complessità	Entusiasmo Ansia per soldi spesi	Allegria Noia per ripetitività Frustrazione	Soddisfazione Ansia da complessità
OPPORTUNITA'	Tutto su una piattaforma Mete Itrate	Biglietti/prenotazioni sotto controllo	Tutto su una piattaforma,attività con info complete, recensioni e foto	Calendario attività semplice, uido con modi ca rapida Incertezze e alternative

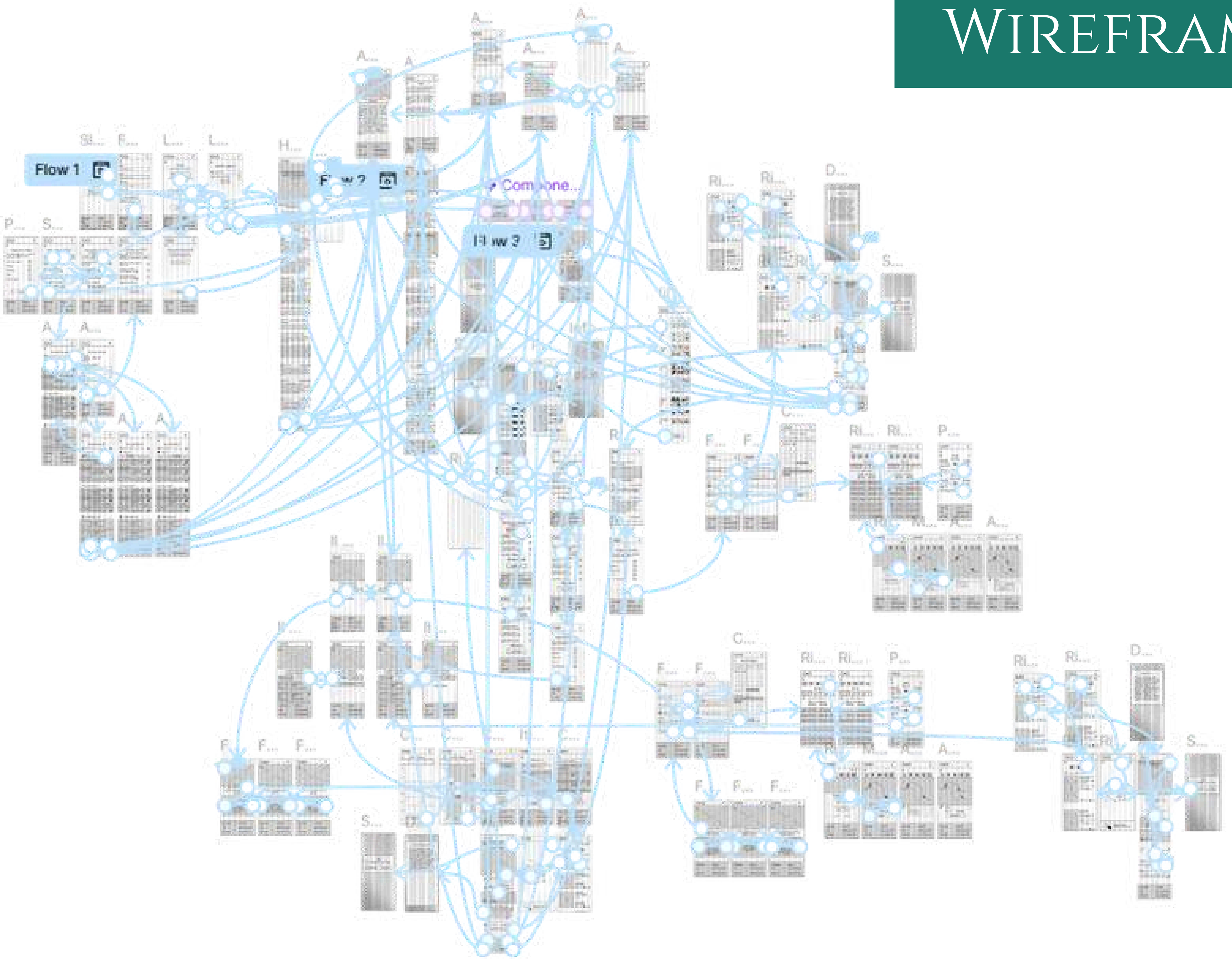
Problems	Chronic indecision	Lack of knowledge	Fragmented transport /lodging search	Planning effort
Solutions	Smart search with filters	Omni-lodging/ transportation	Itinerary builder	
Customer Segments	Particular but undecided travelers	Busy professionals	Reluctant planners	
Unique Value Proposition	The best vacation is the one organized by you, without the stress of doing alone			
Unfair advantage	Algorithmic Integration	Global multi-modal lodging/transport	AI-assisted user flow	
Channels	Seamless conversion path leading users from initial destination search to final booking.			
Revenue Streams	Transactional model	Subscription model	Affiliate fees	
Key metrics	Usage Volume	Average time	Number of saved itinerary/places	Friction rate
Cost structure	Fixed Costs: Domain registration, cloud hosting, APIs licensing	Variable Costs: AI, Pdf, customer support		

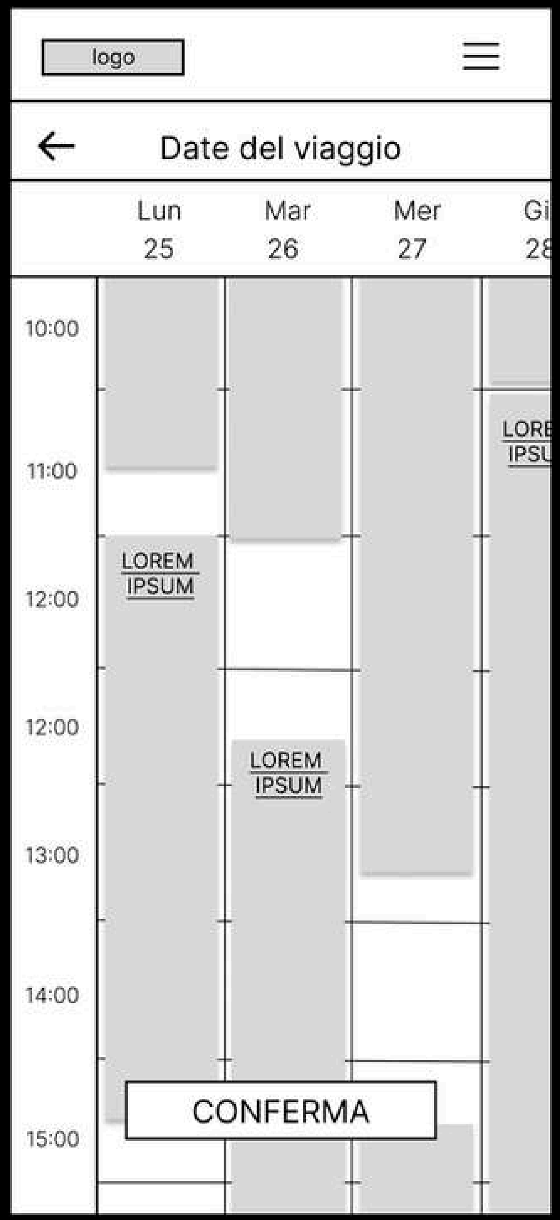
Info architecture



Ux development

Wireframe





[Click here for dekstop prototype]

[Click here for mobile prototype]

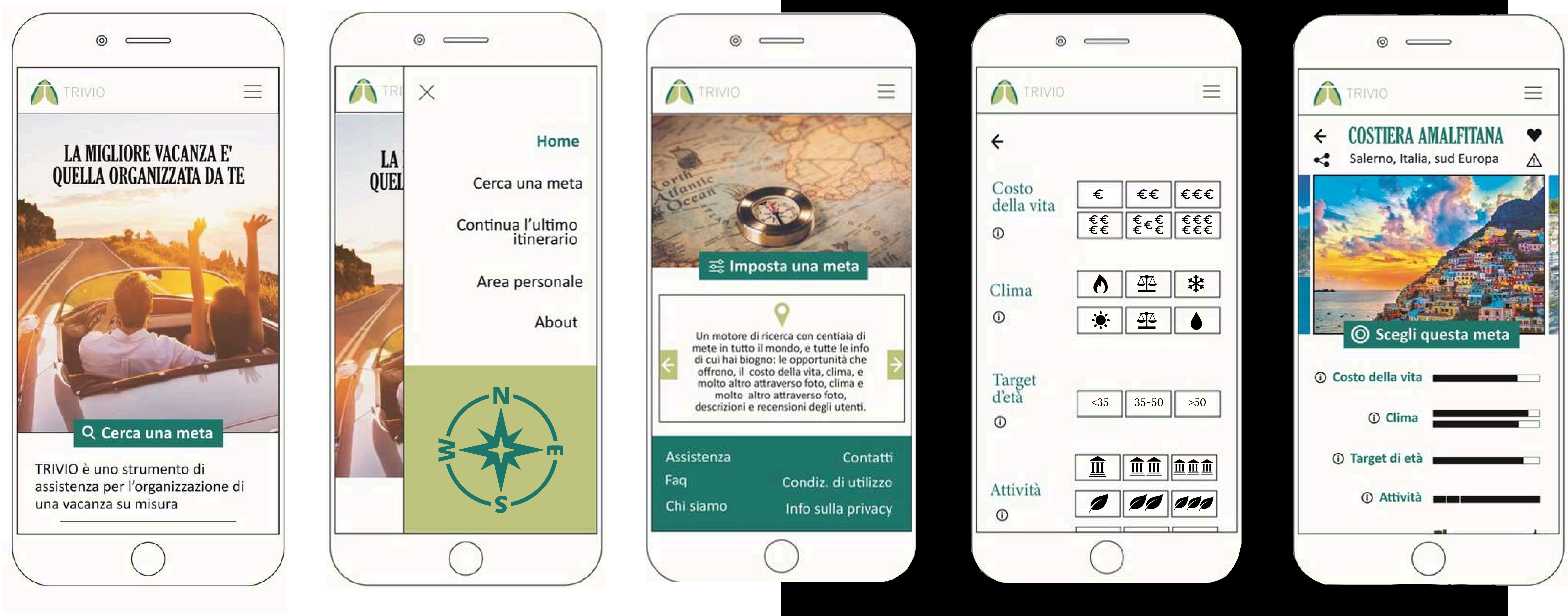
Ui development

Moodboard

Style tile

Ui development

Mock-up



ZeroWaste

A moka funnel

This project stems from a personal frustration: everyone hates spilling coffee grounds all over the counter or floor instead of getting them all into the Moka pot.
This is a funnelt that make all the coffee enter, avoiding waste and disorder.

Analysis

Competitor

Pros

No waste; you can pour the excess coffee back if you put too much.

Cons

It isn't transparent, doesn't provide feedback, requires an extra step to use, isn't universal for different Moka pot sizes.

Pros

Zero waste; it is transparent and provides visual feedback on the amount of coffee.

Cons

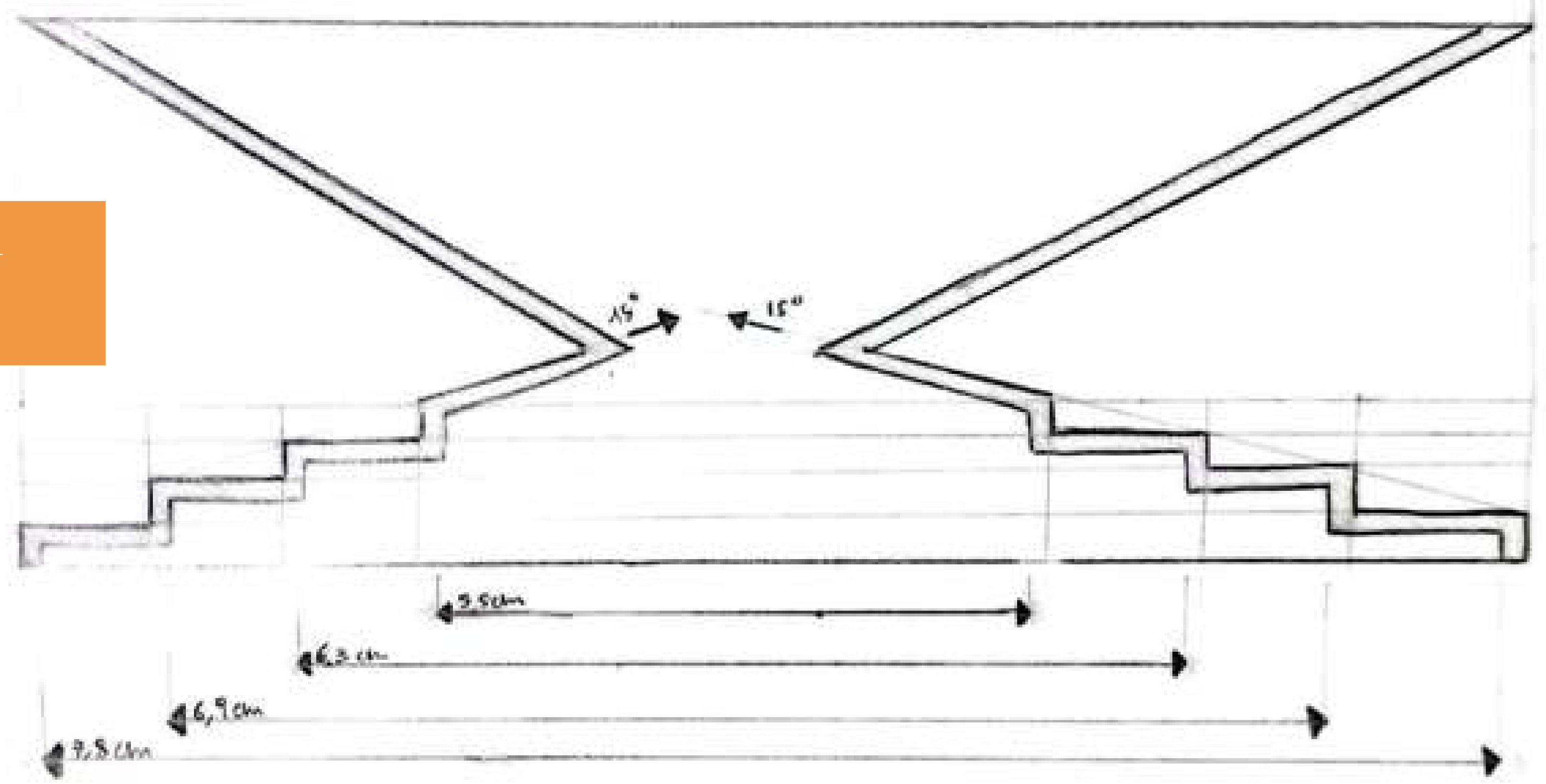
Storyboard

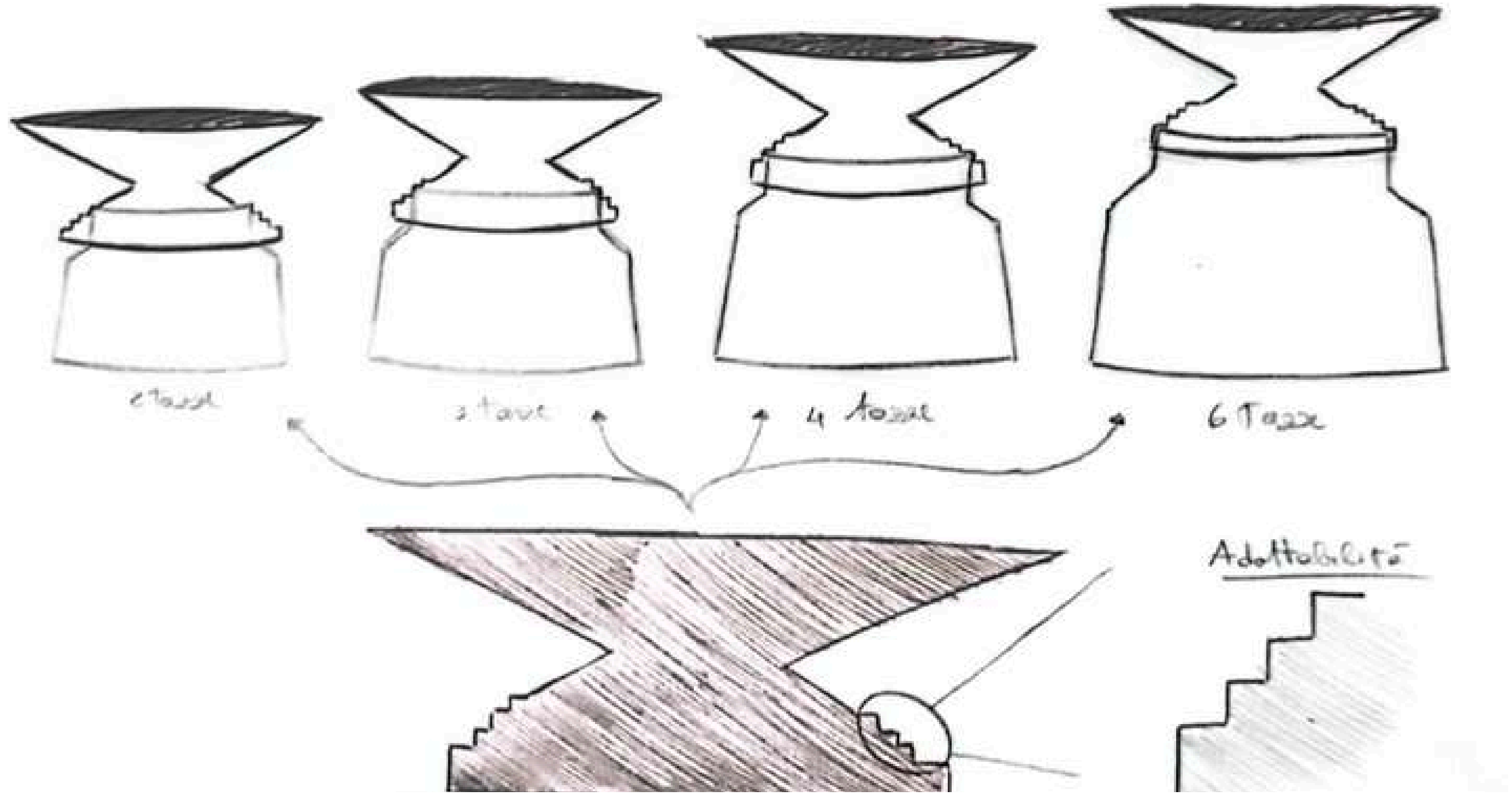
Development

Draft

Adaptability

The prototype draft has been made to understand the right shape and inclinations. Then, the various sizes of the moka pots were studied in order to design the stepped profile in the lower part, so that the funnel could fit all the moka pots.

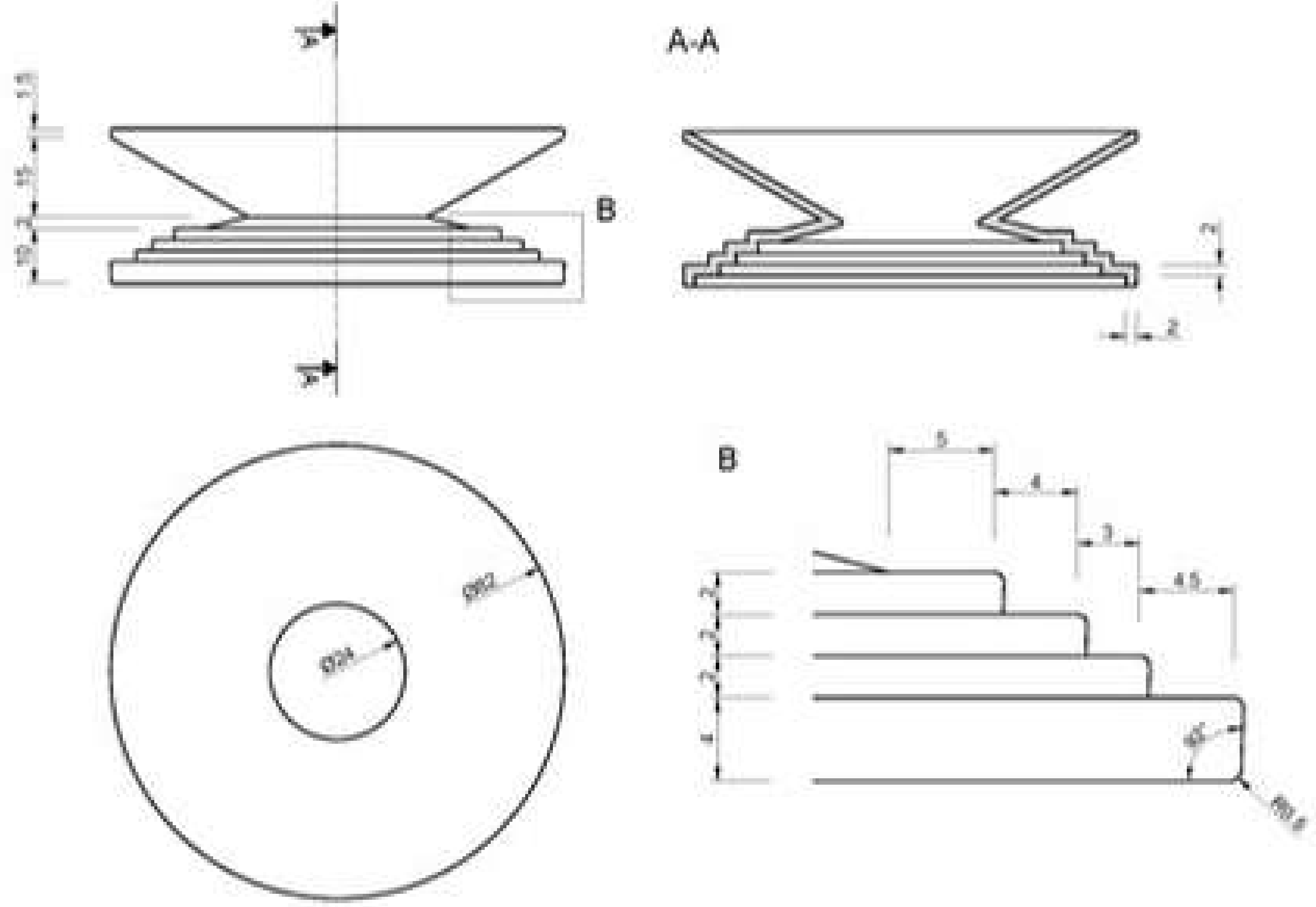




Development

Td

The object has been designed in recycled/bio-based transparent PP, produced by injection molding, with draft angles therefore incorporated.



Rendering

To be continued...